A FIRST-PRINCIPLES GUIDE

Post-Labor Economics

What happens to the economy when machines can do all the mental and physical work

An argument built step by step from axioms,
six scenarios for 2026–2036, and a best guess

November 2026

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ABOUT THIS GUIDE

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The research notes, the data, the models and the code that draws every figure are in the repository:

https://github.com/lastmjs/post-labor-economics

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Nothing here is investment, legal or tax advice. It is an argument about the economy as a whole, and forecasts about the economy are often wrong.

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CHAPTER 1

The question, and the answer in brief

This guide answers one question:

> *Once artificial intelligence automates mental and physical labor entirely, what happens to our economy?*

It is an easy question to answer badly, and most answers go wrong in one of three ways.

The first is to reason from the past without knowing why the past turned out as it did. Machines have replaced human work for two hundred years and workers became richer, so they will again. That is a pattern, and patterns hold until the conditions behind them change. Chapter 3 finds the conditions.

The second is to confuse making things with sharing them out. People say that if nobody has a wage nobody can buy anything, so the economy will collapse, or automation will stop itself. Neither follows. An economy can produce in abundance and still leave most people with no claim on the product. Chapter 7 shows why these are separate matters.

The third is to confuse what machines can do with what they are being used to do. One camp looks at the capabilities of AI and announces that everything changes next year. The other looks at the unemployment rate and announces that nothing is happening. Chapter 11 shows that both are describing the same world.

How the guide is built

The guide is meant to be checked as well as read. It is built as a chain, and the links are numbered.

- An **axiom** is a basic fact about how economies work, stated with the evidence for it. There are nine.
- A **definition** fixes the meaning of a term.
- A **proposition** is a conclusion. Each one names the earlier statements it rests on, so that a reader who doubts it can see exactly which link to test.
- A **forecast** is a judgment about the future, with a probability attached. Forecasts appear only in Part IV and are kept apart from everything else.

Statements are referred to by letter and number: A3 is the third axiom, P12 the twelfth proposition. In the electronic version each reference is a link. Every chapter ends with a box that says where the chain stands, and Appendix B lists the whole chain on two pages.

The guide has four parts. Part I sets out how a market economy distributes what it produces, and why earlier automation raised wages. Part II assumes that machines can do all work and derives what happens to wages, to production and to the distribution of income. Part III asks who has the power to change that distribution. Part IV leaves the assumption behind, looks at

the evidence as of November 2026, builds six scenarios for the next ten years and ends with my own best guess.

The answer in brief

For readers who want the conclusion before the argument, here it is. Every sentence below is defended in the chapter named.

1. In a market economy, people receive a share of what is produced by owning something that production needs. Most people own little except their capacity to work, so wages are how most people get their share (Chapter 2).
2. Machines made workers richer for two centuries because every machine was narrow. There was always something left that only people could do, and whatever is scarce is well paid (Chapter 3).
3. A machine that can do everything a person can do removes that protection. The wage for any work then cannot exceed what it costs to have the machine do it, and that cost keeps falling. This happened once before, to horses (Chapter 5).
4. The effect is not gradual. While any necessary task still needs people, the people who do it can prosper. When none does, the gain reverses. "Nearly all" and "all" are different worlds (Chapter 5).
5. Production does not suffer. An economy with machine labor can produce far more than ours, and grow faster (Chapter 6).
6. Nor does it run out of customers. The income that used to be paid as wages is paid to owners, and production turns to serve them and to build more machines. The economy keeps running. It stops providing for most people (Chapter 7).
7. Some paid work remains wherever a person is wanted as a person. It reflects how income is distributed and cannot repair it (Chapter 8).
8. Majorities have had a share of output because they were needed as workers, soldiers and taxpayers. Automation removes that leverage over time, so any lasting claim on the output of machines has to be secured during the transition (Chapter 9).
9. Providing for everyone would cost a small fraction of what such an economy produces. Claims held as property last; claims that must be voted again each year do not (Chapter 10).

That is the answer to the question as asked. Its short form is: **the economy does not break; the wage does, and the wage was doing more than anyone noticed.**

The forecast for the coming decade is more uncertain and is argued in Part IV. In short: machines will probably be able to do nearly all screen-based work around the end of this decade, and will take a further ten years or so to be put fully to that use. Robots capable of general physical work will follow roughly a decade behind. The change will show up first as a collapse in hiring, then as a recession from which office jobs do not return. By 2036 the United States will most likely be part of the way through: machines doing perhaps a quarter of all work, employment and labor's share of income each down by several points, policy a patchwork of dividends, public stakes and protected jobs, and the question of who owns the machine economy unsettled. I give that central scenario two chances in five, faster ones one in four, slower ones a little under one in four, and a well-managed one about one in eight.

What the guide leaves out

It is about the economy of the United States, with other countries mentioned in passing. It treats the dangers of advanced AI that are not economic, such as loss of human control over the systems or their use as weapons, only where they bear on the argument. They may well matter more than anything discussed here. And it is written at one date. The facts in Part IV are those of 12 November 2026, and some will be out of date within months.

PART I

The machine we live inside

Before asking what breaks, we need to know how the thing works: what an economy is, how people get their share of it, and why two centuries of machines made workers richer and not poorer.

CHAPTER 2

What an economy is

An economy is a way of answering two questions: what gets made, and who gets it. Money, firms, jobs, banks and markets are machinery for answering those two questions. The premise of this guide, machines that can do all the work, makes the first question easier than it has ever been and throws the second one wide open.

To see why, we need a handful of facts about how the machinery works. This chapter sets out five axioms and draws three propositions from them. None of them is controversial among economists. The work lies in keeping all of them in view at once, because the argument of the later chapters is nothing more than these facts applied to a new situation.

Wants, production and ownership

> **AXIOM 1** **People want more than exists**
> At a price of zero, people would take more food, housing, medical care, travel, security and free time than can be supplied. Because wants exceed supply, every society needs some rule that decides who gets what.

Particular wants can be satisfied. Nobody wants a tenth meal in a day. But total wants have never been satisfied in any society on record. People who have enough food want better housing; people with good housing want health, schooling, travel and standing among their neighbors. Some wants cannot be satisfied for everyone even in principle, because they are about rank: the best plot of land, the most admired school, the front row. This matters later, because it means that no amount of production makes the question of distribution go away.

> **AXIOM 2** **Everything is made from four kinds of input**
> Every good and service is produced by combining:
> - **work**: human time, effort and skill;
> - **tools**: machines, buildings, vehicles and software, which are things made in order to make other things, and which economists call capital;
> - **nature**: land, energy and raw materials;
> - **knowledge**: the recipes for combining the other three.

The four inputs differ in one way that will turn out to decide almost everything. Tools can be manufactured: if more are wanted, more can be built. Knowledge, once found, can be used by everyone at once without running out. Nature is fixed in quantity or expands slowly. And work has been limited by the number of people and the hours in their days. An economy can double its stock of machines in a decade. It cannot double its stock of adults.

AXIOM 3 **Every input has an owner**

In a market economy, each input belongs to someone who decides how it is used and collects what it earns. A person's work belongs to that person. Nobody can be made to hand over what they own, except to the state through taxes.

DEFINITION 1 **Income**

A person's income is their claim on what the economy produces. In a market economy a claim can arise in only three ways: by **selling work** (wages), by **owning** something that earns a return (profits, interest, rent, royalties), or by **receiving a transfer** from someone who has a claim (family, charity or the state).

There is no fourth way. Every scheme discussed in this guide, from a universal basic income to a public wealth fund to a law that reserves certain jobs for humans, works through one of these three channels.

PROPOSITION 1 **All output is somebody's income**

Every dollar of output is paid to the owner of some input that helped produce it. The total income of a country therefore equals its total output, and the question "who gets the output?" is the same question as "who owns the inputs that production uses, and what is each input paid?"

Rests on: A2, A3

Follow the money from a single sale. A buyer pays $30,000 for a car. The carmaker pays some of it to its workers, some to its lenders, and some to suppliers of steel, glass and electricity; it keeps the remainder as profit, which belongs to its shareholders. Each supplier does the same with its part. Trace every branch to the end and the whole $30,000 has landed in someone's pocket as a wage, as interest, as rent or as profit, with the government taking a share along the way. Nothing is left over, and nothing is paid out that was not first earned by a sale. National statistics are built on this fact: gross domestic product and gross domestic income are two measurements of the same number.

This identity will do a great deal of work in Chapter 7. It means, for example, that wages cannot fall across an economy without the same dollars showing up as somebody else's income, so long as output itself does not fall.

What a price is

AXIOM 4 **Nobody pays more than they must**

A buyer who can get the same result in two ways takes the cheaper one. A firm that can have a task done by either of two methods of equal quality uses the method that costs less; a firm that does not is undersold by a competitor that does.

PROPOSITION 2 An input is paid for its scarcity, not its importance
The price of an input can never be more than the cost of the cheapest substitute that does the same job. Where there is no substitute, competition among buyers pushes the price up to the value of what one more unit of the input adds. In both cases, what sets the price is how hard it is to get one more unit, not how much the input matters in total.
Rests on: A4

This is the old puzzle of water and diamonds. Water is necessary for life and costs almost nothing. Diamonds are ornaments and cost a great deal. The puzzle dissolves once you notice that a price attaches to one more unit, not to the whole supply. One more gallon of water is easy to get; one more diamond is not.

Apply the same reasoning to work. A wage is the price of an hour of someone's work. It is high when what that hour adds cannot be had more cheaply in any other way, and it is low when it can. A wage is not a measure of effort, of merit or of how much the worker needs the money, and it is not a measure of how important the work is. Home health aides do work on which lives depend and are among the lowest-paid workers in the United States, because many people are able and willing to do that work. **A wage is a price, and it is governed by what governs every other price.** Keep hold of this sentence. Most confusion about automation comes from letting go of it.

How most people get their share

AXIOM 5 Most people own little except their capacity to work
In the United States the less wealthy half of households owns about 2.5 percent of all household wealth, and most of that is the house they live in. The wealthiest tenth owns about two thirds of all wealth and nearly nine tenths of all shares in companies. For a typical household, wages and the benefits that depend on wages are the main source of income for the whole of working life.

PROPOSITION 3 The wage system is the distribution system
Because most people own little except their work, most people's claim on output is their wage. The labor market is therefore doing two jobs at once. It is a **pricing** system, which tells firms where human work is worth using. And it is the **distribution** system, which decides how much of the output most households can buy.
Rests on: D1, P1, A5

The second job goes deeper than paychecks. In the United States most of the safety net is itself built on wages. Social Security and Medicare are financed by a tax on payrolls, and their benefits are earned through years of covered work. Unemployment insurance pays in proportion to past wages. About half the population gets health insurance through an employer. Mortgages are granted against expected wages. Even the federal government's own income rests mainly on work: taxes on wages and salaries supply most of its revenue.

The two jobs of the labor market have gone together for so long that they look like one thing. They are two. One is a matter of engineering and cost: is a person the cheapest way to get this task done? The other is a matter of how people live. For two hundred years the answer to the

first has been "yes, for a great many tasks," and so the second took care of itself. The rest of this guide asks what happens when the answer to the first question becomes "no."

WHERE THE CHAIN STANDS

- Output and income are the same total (**P1**). Whoever owns the inputs that production needs receives the output.
- Any input, human work included, is paid according to how hard it is to replace at the margin, and never more than its cheapest substitute costs (**P2**).
- Most people own only their work (**A5**), so the wage is how most people receive their share (**P3**).
- It follows that anything that makes human work easy to replace is a threat to distribution, even if it is a gift to production. To judge that threat we first have to understand why two centuries of machines did not carry it out.

CHAPTER 3

Why two centuries of machines made workers richer

Machines have been taking over human work since the 1700s, and over that time the pay of ordinary workers in the industrial countries has risen more than tenfold. Anyone who predicts that the next machines will be different has to explain first why the earlier ones were not. This chapter gives that explanation, and gives it in a form precise enough to test against the premise of this guide.

The standard reassurance goes like this. In 1800 about three quarters of Americans worked on farms. Today fewer than two in a hundred do, and the country grows far more food. The farm workers were not left idle forever; their grandchildren became machinists, clerks, nurses and programmers. People who warned that machines would cause permanent mass unemployment, from the Luddites in 1811 to the authors of the "Triple Revolution" memorandum sent to President Johnson in 1964, were wrong each time. Economists named the mistake the "lump of labor fallacy": the belief that there is a fixed amount of work to be done, so that a machine which does some of it leaves less for people.

The reassurance is correct about the past. But "it turned out fine before" is an observation, and what we need is the mechanism. A mechanism has conditions, and conditions can stop holding.

Machines take tasks, and tasks are complements

> DEFINITION 2 **Task**
> A task is one distinct piece of work needed to produce something: welding a seam, diagnosing a fault, answering a customer's question. A job is a bundle of tasks. A machine never takes over a job as such. It takes over tasks.

> AXIOM 6 **Most production needs many tasks, all of them**
> Making almost anything requires a number of different tasks, and more of one does not make up for the lack of another. A car needs designing, stamping, welding, wiring, testing, shipping and selling. Twice as much welding does not make up for no wiring. In the language of economics, tasks are *complements*.

The evidence for this axiom is all around. It is why a strike by a few hundred port crane operators can stop the trade of a whole coast. It is also the reason behind a famous pattern called Baumol's cost disease: the parts of the economy where productivity has grown slowly, such as health care and education, take a growing share of total spending, exactly because the things that became cheap could not be used in place of the things that did not.

Put the last two statements together with the rule about prices from the previous chapter and a result follows at once.

PROPOSITION 4 The step that machines cannot do collects the reward

When machines make some tasks cheap and plentiful, the remaining tasks become the scarce step in production. Since inputs are paid for scarcity, more of the value of the product flows to whoever performs the remaining tasks. If people are the only ones who can perform them, automation raises wages.

Rests on: P2, D2, A6

A WORKED EXAMPLE: THE SHIRT ECONOMY

A hundred workers make shirts. Each shirt needs one unit of cutting and one unit of sewing, and a worker can do one unit of either in an hour. Fifty workers cut and fifty sew, so the economy makes 50 shirts an hour. The workers are the only input, so they receive all 50 shirts between them: **half a shirt per hour of work**.

Now someone invents a machine that cuts cloth for next to nothing. All hundred workers move to sewing, and the economy makes 100 shirts an hour. Sewing is now the only scarce step, so the sewers collect nearly all the value: close to **one shirt per hour of work**. The machine took over half of all the tasks that workers used to do, and the workers' real pay doubled.

Notice what the result required. Sewing still had to be done by a person. The cutters had to be able to become sewers. And somebody had to want the extra 50 shirts. We will return to this economy in Chapter 5 and add a second machine.

The three conditions

The shirt example shows the mechanism in miniature, and it shows the conditions on which the mechanism depends. Stated generally:

PROPOSITION 5 Why past automation raised wages

Automation has raised the pay of workers when, and because, three conditions held together.

1. **Something was left.** There remained tasks that only people could do.
2. **What was left was needed.** Those tasks were complements to what the machines produced, so demand for them grew as machine output grew.
3. **People could get there.** Most workers, or their children, could move into the remaining tasks.

Rests on: P4, A1

For two hundred years the first condition held for a simple reason. Every machine ever built was *narrow*. A power loom weaves and does nothing else. A tractor pulls. A spreadsheet adds. A person, by contrast, is a general-purpose system: the same body and brain can weave, pull, add, negotiate, comfort a child and learn a trade that did not exist the year before. Whenever a narrow machine took over one task, there were always other tasks for which a general-purpose worker was the only candidate.

> **AXIOM 7 Until now, people could do things that no machine could**
> Through every earlier wave of automation, people kept abilities that no machine had: flexible reasoning, language, fine control of the hands in cluttered surroundings, and the understanding of other people. Each wave of machines pushed human work toward the abilities that remained exclusive.

You can read economic history as a sequence of such moves. Engines replaced muscle, and people moved to work that needed dexterity and attention on the factory line. Factory machines and then industrial robots took over the repetitive movements, and people moved to offices, to work that needed reading, calculating and coordinating. Computers took over the routine calculating and filing, and people moved again, toward work that needed judgment, persuasion, creativity and care, and toward manual work in unpredictable places, such as plumbing a hundred-year-old house. One study found that about 60 percent of American workers in 2018 held kinds of jobs that did not yet exist in 1940. The new jobs appeared, and people filled them, because people were the only general-purpose workers there were.

The second condition held because of **A6** and **A1**: the remaining tasks were needed, and there was no limit to what people wanted. The third held most of the time, though slowly and at real cost.

When the third condition failed

It is worth pausing on that cost, because history's reassurance is weaker than it sounds even for the episodes that ended well.

In Britain between about 1800 and 1840, output per worker rose by nearly half while real wages hardly moved. The gains went to the owners of the new factories. Only after about 1840 did wages begin to rise in step with productivity. The historian Robert Allen named these decades "Engels' pause," after Friedrich Engels, who described the condition of English workers in 1845 as the pause was ending. A worker who was twenty when it began was sixty when it ended.

The handloom weavers of England are the sharpest case. There were roughly a quarter of a million of them in the 1820s, skilled workers with a respected trade. The power loom did not make them unemployed at a stroke. It did something slower: it put a falling ceiling on what a piece of handwoven cloth could sell for. Their earnings fell by more than half over about three decades, and the trade had all but vanished by the 1860s. Few of the weavers themselves moved to anything better. The gains from the loom were real and eventually reached almost everyone, but they reached a different generation.

The more recent case is the arrival of Chinese manufactured imports in the United States after 2000. Economic theory predicted that workers in the affected towns would move to other industries or other places. Mostly they did not. More than a decade later the towns that had lost the most factory work still had lower employment, lower wages, more dependence on disability payments, fewer marriages and more deaths from drugs and alcohol.

The lesson to carry forward is that even when all three conditions hold for the economy as a whole, adjustment takes decades and its costs fall on particular people who are rarely compensated. That is the *good* case.

The horse

There is one large group of workers in modern history for whom the first condition failed completely. They were not human, which is why they are usually left out of the reassuring story, but the economics is exact.

In 1915 there were about 26 million horses and mules at work in the United States, pulling plows, wagons, streetcars and artillery. The horse had already survived one great wave of machines. Railways took over long-distance haulage in the 1800s and the number of working horses *rose*, because every ton that arrived by rail still had to be carted from the station. That was **P4** at work: the horse did the step the railway could not.

Then came the internal combustion engine, in the form of the tractor and the truck, and this machine was different in one respect. It could do everything a horse could do. No remaining step in farming or haulage needed a horse. From that point the price of a horse's work was capped by the cost of getting the same pulling and carrying from an engine (**P2**), and that cost fell every year. A horse, however, has an upkeep that does not fall: several acres' worth of feed, a stable, a farrier and a vet. Once the value of a horse's work dropped below the cost of keeping a horse, it no longer paid to breed one. By 1960 the country had about 3 million horses, and most of those were kept for pleasure.

> **PROPOSITION 6 The horse lemma**
> Suppose that for *every* task an input can perform there is a substitute that performs it at lower cost than the input's own upkeep. Then the price of the input falls below its upkeep, and its use in production falls toward zero. Total output rises while this happens.
> Rests on: **P2**

Nothing about this required horses to become worse at their work. A horse in 1950 was as strong and as willing as a horse in 1900. Nor did the economy have less need of pulling and carrying; it did far more of both. The work remained, and the horse stopped being the cheapest way to get it done.

The economist Wassily Leontief, who won the Nobel prize for mapping how industries feed into one another, drew the comparison in 1983: the role of humans as the most important input to production, he wrote, was bound to shrink in the same way the role of horses had. For forty years this was treated as a provocation, because the first condition of **P5** so obviously held. Machines were narrow.

Three ways people are not horses

The comparison is uncomfortable, and it should be handled with care, because there are three real differences between a population of horses and a population of people. Horses were never the purpose of the economy; people are. Horses owned nothing; people own things. And horses had no say in what happened to them; people vote.

Put in terms of this guide: people are the *customers* for whom production exists, they are (some of them) the *owners* of the machines, and they are the *citizens* of states that can tax and redistribute. Each of the three is a possible way out of the horse's fate. Whether any of them works is not settled by economics alone, and Parts II and III take them one at a time. For now it

is enough to state what the horse does establish: an input's long record of benefiting from new technology is no protection once a technology arrives that can do everything the input does.

WHERE THE CHAIN STANDS

- Machines take tasks. Because tasks are complements, the tasks that machines cannot do become the bottleneck and collect the reward (**P4**).
- Automation raised wages for two centuries because three conditions held: something was left for people to do, it was needed, and people could move into it (**P5**). The first condition held because machines were narrow and people were general (**A7**).
- Even then, adjustment took a generation or more and the losers were seldom made whole.
- Where the first condition fails entirely, the horse lemma applies (**P6**): the input's price falls below its upkeep while output grows.
- So the whole question turns on the first condition. The next chapter looks at a machine designed to remove it.

PART II

The premise and what follows

Suppose machines can do every task that people are paid to do. What happens to wages, to production, and to the distribution of what is produced? These chapters reason from the premise alone.

CHAPTER 4

The premise: a machine that can do the work

Part I ended on a condition. Automation has made workers richer for two centuries because something was always left that only people could do, and the reason something was always left is that every machine was narrow. This chapter states the premise that the rest of the guide examines: a machine that is not narrow.

What is being assumed

> **DEFINITION 3** **Machine labor**
> Machine labor is hardware and software that performs a task to the standard of a person who is competent at that task, without a person doing part of it. *Mental* machine labor needs only a computer and a connection. *Physical* machine labor also needs a body: a vehicle, an arm, a robot.

> **DEFINITION 4** **The premise**
> Machines can perform every task that people perform for pay, at least as well as people do, and at a lower cost than the cost of supporting a person.

Parts II and III take the premise as *completely* true and work out what follows. That is a deliberate choice of method. It is the only way to see the destination clearly, and several of the results turn out to depend sharply on the difference between "nearly all tasks" and "all tasks." It is not a claim that the premise is true today, and a reader who doubts that it ever will be should read those parts as an extended "if." Part IV then asks the empirical questions: how much of the premise is already true, how fast the rest is arriving, and what is likely to happen along the way.

Why this machine is different in kind

A power loom was designed to weave. Nobody expected it to learn bookkeeping. Each earlier machine was built for a purpose, and its builders had to understand the task well enough to mechanize it.

The systems now being built are made differently. They are trained on the accumulated record of what people have written, said, drawn, coded and done, with the aim of producing whatever a capable person would produce in the same situation. Nobody programs the skills in one by one. A system trained this way that learns to write contracts also turns out to write software, summarize medical studies and plan a wedding, because what it has absorbed is the general pattern of competent human behavior. The property that kept people in work through every earlier wave of automation was generality (A7), and generality is the property this technology is designed to reproduce.

That is why the reassuring history of Chapter 3 cannot simply be extended forward. The history is a record of what happens when narrow machines meet general workers. It contains only one case of a general substitute meeting the workers it could fully replace, and those workers were horses.

Five properties of machine labor

Suppose such a machine exists. As an input to production it differs from human work in five ways, and each one matters later.

It is owned. A machine worker is a piece of capital. It is built, bought and sold, and what it earns belongs to its owner (A3). A human worker owns their own work and is paid for it. This single difference is the root of the whole distribution problem.

It can be copied. A person takes about twenty years to raise and educate, and each one has to be trained separately. When one machine has learned a skill, the skill can be copied to a million machines for the cost of the copying. So the supply of a newly valuable skill, which among people is scarce for a generation, can be expanded among machines almost at once.

It gets cheaper. The price of a fixed level of AI capability has been falling roughly tenfold a year, through better chips, better methods and competition among suppliers [1]. Nothing about the cost of a person's upkeep falls like that.

It does not rest. A machine can work all 168 hours of the week. A full-time person works about 40.

It can work on itself. Machines that can do every task can do the tasks of building machines and of improving their design. The stock of machine labor can then grow by its own effort, which no stock of tools has been able to do before.

How much of the premise is true today

The detailed evidence is in Chapter 11. In outline, as of November 2026:

For *mental* work, the premise is partly true and is advancing quickly. The best AI systems can now carry out, unattended, software projects that would take a skilled person days. One independent measure tracks the length of task that systems complete half the time: it stood at one hour in early 2025 and at seventeen hours in spring 2026 [2]. On a benchmark made of real freelance projects graded against the paid professional's work, the best system completed 2.5 percent of projects in October 2025 and 41 percent in November 2026 [3]. But on day-long professional tasks in finance and law the pass rate is still close to zero, about half the code that passes automated tests would be rejected by the people who maintain the software, and the systems do not get better at a job by doing it, as every human employee does [4], [5].

For *physical* work, the premise is true in one large task and barely true elsewhere. Driving has been automated in the sense that matters: one company carries half a million paying passengers a week with no driver, with a much better safety record than human drivers. General-purpose robots are at a far earlier stage. The best published result has a robot completing 56 percent of long household chores in homes it had never seen, working at a fraction of human speed. Fewer than a hundred thousand humanoid robots exist, and most are in laboratories [6].

One caution applies to all such figures. Completing a task half the time is not the same as being employable. An employer needs something much closer to always, and the length of task that current systems complete four times in five is about one fifth of the length they complete half the time. Much of the distance between today's systems and the premise lies in that gap between usually and reliably.

WHERE THE CHAIN STANDS

- The premise is that machines can do every paid task at least as well as people, for less than it costs to support a person (**D4**).
- It differs from all earlier automation because the machine is general, which removes the condition that protected workers before.
- Machine labor is owned, can be copied, gets cheaper, works around the clock and can build more of itself.
- Today the premise is partly true for mental work, true for driving, and mostly untrue for other physical work.
- Part II now assumes the premise holds in full, and asks what happens to wages, to production and to the distribution of income.

CHAPTER 5

What happens to the price of work

This chapter is the center of the argument. It takes the premise of Chapter 4, a machine that can do the work, and the rule about prices from Chapter 2, and derives what happens to wages. The derivation has four steps. Machines put a ceiling on the wage. The ceiling tends toward the cost of making and running the machine. While some tasks still need people, wages can rise, but when none do they fall to the ceiling. And the ceiling is lower than the cost of keeping a person.

Step one: the ceiling

> **PROPOSITION 7** **The wage ceiling**
> In any task that a machine can do as well as a person, no employer will pay a person more than it costs to have the machine do it. The price of the machine's work is a ceiling on the wage for that task.
> Rests on: A4, P2, D3

This is P2 with the word "input" replaced by "work." It needs no assumption about greed or about the intentions of any company. An employer who paid people more than the machine price for the same result would have higher costs than a competitor who did not, and would lose customers to that competitor (A4).

Return to the shirt economy of Chapter 3. After the cutting machine arrived, a hundred workers sewed, and each earned about one shirt per hour of work. Now add a second invention.

> **THE SHIRT ECONOMY, CONTINUED**
>
> Someone invents a sewing machine that needs no operator. Building and running it uses up resources worth one fifth of a shirt for every shirt it sews.
>
> A human sewer still sews one shirt an hour, as well as ever. But a shirt's worth of sewing can now be bought from the machine for a fifth of a shirt, so nobody will pay a person more than that for it. The sewer's pay falls from one shirt an hour to **a fifth of a shirt an hour**, a cut of 80 percent.
>
> Meanwhile the economy can make as many shirts as it has machines. Output is no longer limited by its hundred workers at all.
>
> The first machine doubled the workers' pay. The second cut it by four fifths. They were the same kind of invention in the same economy. The only difference is that after the first machine there was still a task that only people could do, and after the second there was not.

Step two: what sets the price of machine work

The ceiling is only as low as the price of machine work. So what determines that price?

PROPOSITION 8 Machine work is supplied at cost

Machine work is a manufactured product. Whenever its price is above the cost of producing it, including a normal return on the money invested, it pays to build more machines, and the extra supply pushes the price down. Over time the price of machine work tends toward its cost of production, and that cost falls as the technology improves.

Rests on: A2, A4, D3

This is the deepest difference between human work and machine work, and it is easy to miss. The number of people is, for economic purposes, fixed. When demand for work rises, the supply of people cannot respond, so the wage rises instead. That is the entire reason economic growth has made workers richer: capital and knowledge piled up, people did not, and the scarce input was paid more.

Machine work has no such limit. If demand for it rises, factories make more chips and more robots. Supply responds in quantity, and price returns to cost. Figure 1 shows the two cases in one picture. Before, the supply of work is a vertical line at the number of people, and the wage is wherever demand crosses it. After, anyone can add to the supply of work at machine cost, so the supply of work is a horizontal line, and the wage is wherever that line sits.

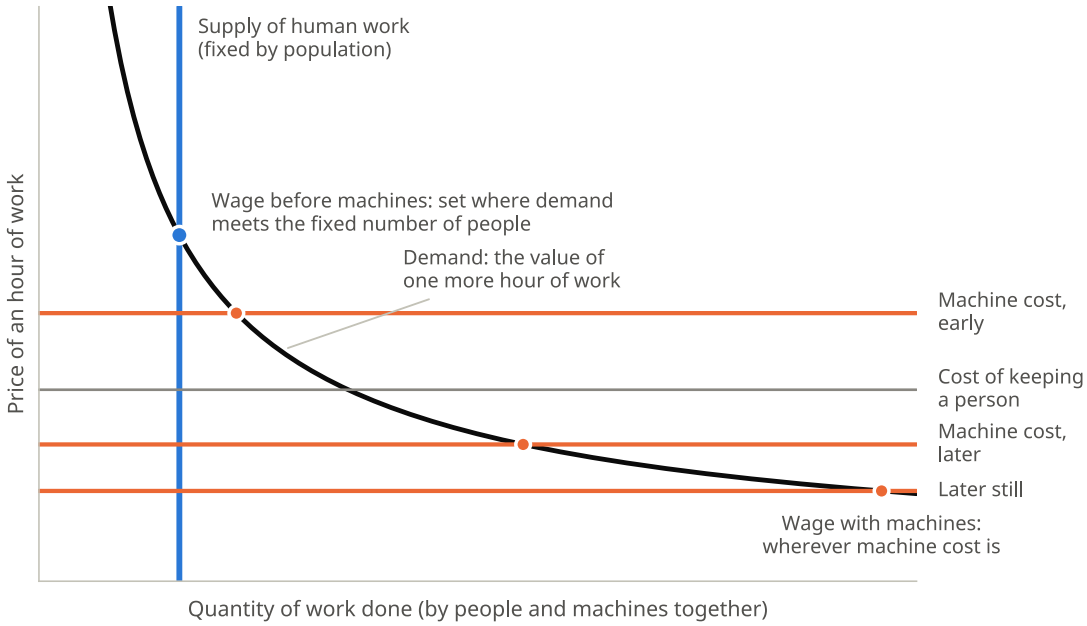


Figure 1. A substitute turns the supply of work from a wall into a floor. When only people can do the work, its price is set by how much work is wanted and how few people there are. When machines can do it, any amount of work is available at machine cost, and that cost becomes the price. As machine cost falls, the wage follows it down the demand curve while the amount of work done grows. The diagram is conceptual and has no units.

One qualification matters a great deal for timing. “Tends toward cost” is a statement about the long run. Building chip factories, power plants and robot plants takes years. While machine capacity is still short of what is wanted, machine work is priced by its scarcity, above its cost,

and wages in the affected tasks can stay high. The speed at which wages fall is therefore set by the speed at which machine capacity can be built. Chapter 12 shows that this speed is very different for mental work and for physical work.

Step three: before the last task, and after

The first two steps apply task by task. A real economy has thousands of kinds of tasks, and machines do not learn them all on the same day. What happens along the way?

We already have the answer for the tasks machines cannot yet do. By **P4**, those tasks become the bottleneck, and the people who do them are paid more. So during a period of partial automation, two forces act on workers at once. Workers in automated tasks hit the ceiling (**P7**). Workers in the remaining tasks gain from scarcity (**P4**), though they are also joined by everyone displaced from the automated tasks, which pulls their pay down again.

To see how these forces add up, I built the smallest model that contains all of them. It has many tasks that are complements, machines that can do a growing share of the tasks and that improve every year, people who move freely to wherever they are still needed, and owners who reinvest a constant share of income. It is a stripped-down version of the model that the economists Anton Korinek and Donghyun Suh published in 2024, and it reproduces their central result [7]. Its equations and the checks on them are in Appendix A, and the code is in the repository. Figure 2 shows two runs of the model. They are identical in every respect but one. In the first run, automation stops when machines can do 90 percent of tasks. In the second it goes on to 100 percent.

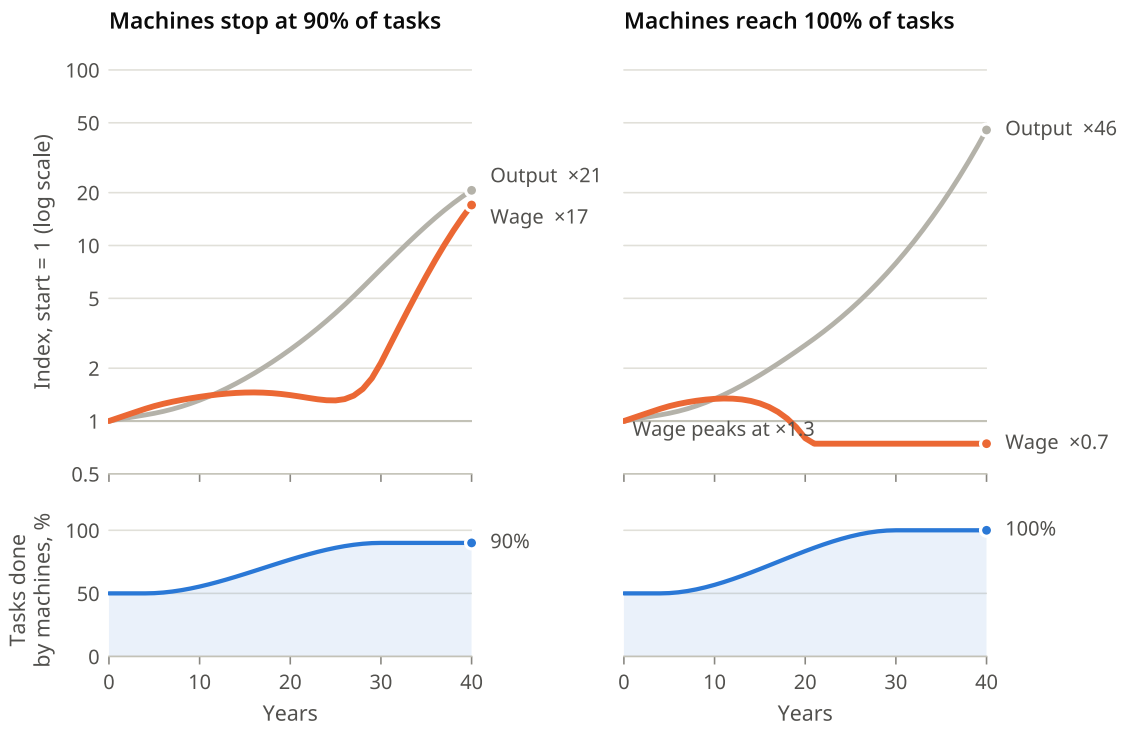


Figure 2. The same economy with and without a last human task. In both runs machines improve by 5 percent a year and the share of tasks they can do rises over 26 years. On the left the share stops at 90 percent: the wage stalls while displaced workers crowd into the remaining tasks, then climbs steeply once those tasks are the bottleneck for a much larger economy. On the right the share reaches 100 percent: the wage peaks about a third above its starting level and then falls to the machine price, 26 percent *below* where it started, while output grows 46-fold. Model output; an illustration of the mechanism and not a forecast.
Source: `models/fig_cliff.py`

PROPOSITION 9 The cliff

The effect of automation on wages changes sign between "nearly all tasks" and "all tasks." While any necessary task remains that only people can do, the people who do it can capture a large share of a growing economy. When no such task remains, that share disappears, and human work is worth what machine work costs.

Rests on: P4, P7

Three things follow from the cliff, and each will matter for the forecasts in Part IV.

The statistics can look healthy until late. On the right-hand side of Figure 2, wages *rise* for the first eleven years of the transition. Anyone watching average wages in that period would conclude, correctly from the data and wrongly about the future, that automation was good for workers.

The premium goes only to those who can do what is left. The model assumes that every worker can move into the remaining tasks. In life, as the set of remaining tasks shrinks, it comes to need rarer abilities, credentials or positions. The bottleneck premium is then collected by fewer and fewer people, and the rest are in the right-hand panel even while the economy as a

whole is still in the left-hand one. For an individual, the question that matters is whether there remains a necessary task that machines cannot do and that *this person* can.

"Almost everything" and "everything" are different worlds. A forecast that machines will do 90 percent of today's tasks and a forecast that they will do all of them sound similar and have opposite implications for workers. Much of the disagreement among economists about AI comes down to this: the optimists expect that some necessary human task will always remain, and the pessimists accept the premise of this guide. Given their respective premises, both camps reason correctly.

Step four: the ceiling is below the cost of living

Suppose, then, that the premise holds completely and there is no last task. Human work is paid the machine price, and the machine price tends to machine cost. How low is that?

The first three steps were logic, and every formal model of full automation agrees with them. This step is different. Economic theory does not say whether the machine price is high or low. That is a question of fact about engineering and costs, and it has to be answered with numbers.

> **PROPOSITION 10** **The upkeep comparison**
>
> A machine that does the work of a person needs fewer resources to build and run than a person needs in order to live. This is already true, at today's prices, for the tasks machines can perform, and the gap widens every year because the cost of machines falls and the needs of a human body do not.
>
> Rests on: P8, D3

Here are the numbers in money.

An adult living alone at the federal poverty line in the United States spends about $15,650 a year. Spread over a full-time working year of 2,080 hours, that is about $7.50 for every hour worked, and it buys a hard life. The median American worker is paid about $24 an hour, to which employers add roughly another 40 percent in benefits and payroll taxes.

Now the machines. A top AI model in late 2026 can be rented for a few dollars per million words of output. Running one continuously on a demanding task for an hour, reading and writing far more than any person could, costs on the order of one to five dollars at list prices, and the price of a given level of capability has been falling by a large factor every year. A general-purpose robot priced at $30,000, lasting five years and working twenty hours a day, costs under a dollar per hour of work to own, plus pennies for electricity and something for repairs.

Neither machine can yet do everything the person can. That is the premise, and whether it will come true is the subject of Part IV. The point here is narrower: *where* a machine can do the work, the price of its work is already a fraction of what it costs to keep a person alive in the United States, let alone what a person is paid.

The same comparison can be made without money, in physical resources, and this version is the one that does not depend on any price. A person needs, at the very least, food, water, shelter and warmth, every hour of every day, whether working or not, for about twenty years before any useful work is done and for all the years after it stops. The energy flowing through the American economy on behalf of each resident is roughly 9,000 watts around the clock. A

working humanoid robot draws a few hundred watts, only while it works. A share of a data-center computer running the mental work of one person draws a few hundred to a thousand watts. The machine is the cheaper way to turn energy and materials into work, by a wide margin, for the same reason that the truck was a cheaper way than the horse to turn them into haulage.

Combine the four steps. The first three give a result that holds in every formal model of full automation that economists have written down.

> **PROPOSITION 11** **The wage stops carrying the output**
> If machines can perform every task that people perform for pay, the wage for human work equals the price of machine work. It no longer rises as the economy grows, and the share of national income paid as wages tends toward zero. This happens while total output rises.
> Rests on: **P7**, **P8**, **P9**

This alone is enough to break the wage system as a way of distributing output (**P3**). Even if wages stayed exactly where they are today in what they can buy, an economy that grew tenfold with a constant wage would be paying its workers a tenth of the share they receive now, and the rest would go elsewhere.

Adding the fourth step gives the stronger result, which is the horse lemma (**P6**) applied to people.

> **PROPOSITION 12** **Work stops paying for life**
> Because machine work costs less than it costs to support a person, the wage under full automation falls below what a person needs to live on. Selling work can then no longer be the way most people earn a living.
> Rests on: **P11**, **P10**, **P6**

I hold the first of these two propositions with near certainty, given the premise. I hold the second with less, perhaps four chances in five, because it depends on magnitudes that the last objection below disputes. The rest of the guide needs only the first.

Notice what neither proposition says. Neither says that there will be nothing for people to do, or that people will be forbidden to work. Anyone may still offer work at the machine price. In practice the result would be recorded mostly as people leaving the labor force, since minimum-wage laws forbid, and few people accept, work at a dollar or two an hour.

Nor does either say that the price of every kind of human work falls to the machine price. Where a buyer wants the work done *by a person*, and a machine is therefore not a substitute at all, **P7** does not apply. Chapter 8 measures how much that exception is worth.

Seven objections

The conclusion is unwelcome, and there are well-known arguments against it. Each deserves its strongest statement.

> OBJECTION **Comparative advantage guarantees people a job**
> David Ricardo showed in 1817 that even if Portugal makes both wine and cloth more efficiently than England, both countries gain when each specializes in what it does *relatively* best. In the same way, even if machines are better than people at everything, machine capacity is limited, so machines should be used where their advantage is greatest and people will be employed in the rest. Some economists conclude that people will keep well-paid work however capable machines become.

The logic of comparative advantage is sound, and it guarantees less than it seems to. It shows that there will be *some* wage at which it pays to employ a person. It does not show what that wage is. In Ricardo's own example the less productive country is the poorer one. What a person is paid for a task under comparative advantage is what it would cost to have the machine do that task, which is **P7** again.

The argument also rests on an assumption that holds for countries and fails for machines. Portugal cannot be copied. Ricardo's theorem takes the quantity of each country's resources as fixed. Machine capacity is manufactured (**P8**). If there is any task for which paying a person is cheaper than building one more machine, the gap is an invitation to build the machine. Comparative advantage therefore protects wages only for as long as machine capacity cannot be expanded, which is to say during the build-out. That period is real, it may last years, and it is part of the scenarios in Part IV. It is a transition, though, and the argument gives no reason to expect it to last.

Finally, the horse had a comparative advantage too. There were always tasks, on steep ground or on small plots, where the tractor's edge over the horse was smallest. That did not keep the value of a horse's work above the cost of its hay.

> OBJECTION **New kinds of jobs will appear, as they always have**
> Most of today's jobs did not exist a century ago. Nobody in 1900 could have predicted the web designer or the MRI technician. It is a failure of imagination to assume that this time no new work will appear.

New tasks will certainly appear. A richer and more complicated economy will want things done that nobody has yet thought of. The question is who will do them. New tasks went to people in the past because people were the only general-purpose learners available (**A7**). Under the premise, there is a second general-purpose learner, and it learns a new task once and then copies the skill to every machine at no cost, while a person takes years to train. The objection assumes that the new tasks will belong to the first of **P5**′s three conditions, "something was left." The premise is exactly that this condition fails.

> OBJECTION **Prices will fall, so even a small wage will buy a great deal**
> If machines make everything cheaply, the cost of living collapses. A wage that looks tiny in today's dollars might buy a comfortable life.

Prices of machine-made things will indeed fall, and this is very good news for anyone who has an income (Chapter 7 returns to it). But it does not rescue the wage, because under the premise

human work is itself priced as a machine-made thing. The wage falls in step with the price of machine work. Goods made entirely by machines fall in price about as fast, so an hour of human work buys no more of them than before. And things that are not made by machines at all, above all land in places where people want to live, become more expensive relative to the wage, because the machine economy and its owners are bidding for them too. A person paid the machine price can afford what a machine needs. People need more than machines do.

> OBJECTION **People working with AI will always beat AI alone**
> The best results come from human judgment combined with machine speed. Machines will be tools that make workers more productive, as computers did.

Where this is true, it describes a remaining human task, namely directing and checking the machine, and **P4** applies: the people who do it will be well paid. It is an accurate description of much work in 2026. But it is a claim about each task at a given moment, and it can stop being true. In chess, teams of a person and a computer beat the best computers for roughly fifteen years after a computer first beat the world champion. Then the person stopped adding anything. The premise of this guide is the situation after that point, for every task.

> OBJECTION **Human wants are unlimited, so there will always be more work to do**
> The "lump of labor" is a fallacy. There is no fixed amount of work. A richer society finds new things to want.

Agreed on every point (**A1**). Unlimited wants guarantee unlimited demand for *output*. They guarantee demand for human work only if human work is the cheapest way to produce the output. The fallacy lies in assuming a fixed amount of work. The argument of this chapter assumes nothing of the kind: in Figure 1 the amount of work done grows enormously. It is done by machines.

> OBJECTION **Workers could always go on as before, so they cannot end up worse off**
> Nothing forces anyone to use the new machines. Workers and the firms that employ them could simply carry on producing and trading with one another exactly as they do today. Since that option always remains, competition can only make them better off. Several formal results in economics say exactly this: under the right conditions, a new technology cannot lower the total income of the inputs that existed before it.

The theorem is correct, and its conditions tell us exactly where to look. The economy of today does not run on work alone. It runs on work combined with factories, offices, land, energy and materials, and the workers do not own those. Their owners are free to put them to whatever use pays best. If land is worth more under a data center than under a factory, and electricity is worth more running machines than lighting an office, then "carrying on as before" is no longer on offer to someone who owns only their work, because the other half of the old arrangement has left. What the theorem really guarantees is that the owners of the old inputs *taken together* do not lose. It is silent on how the gain is divided between the people who supplied the work and the people who supplied everything else. In the most careful recent model of this kind, by Pascual Restrepo, the total paid in wages does not fall when computing power is the only

scarce thing, and yet the wage share of income still goes to zero and wages stop growing with the economy [8]. That is **P11**.

> OBJECTION **Machines will stay scarce, so their work will stay expensive**
> In an economy growing explosively, machines are enormously productive, and the return on owning one is very high. If human work is paid what an equivalent machine earns, and machines earn a great deal, then human wages could stay high or even rise.

This is the most serious objection in economic theory, and several formal models produce exactly this result, for a time or for good [8], [9]. A survey of the theory concludes that whether wages end up above or below today's level is not settled by the models, and depends on how scarce machines remain and on whether land and energy become the binding limit. The reply is therefore about magnitudes. What a person can earn is the rental price of the *amount of machinery needed to replace that person.* For mental work this is a small fraction of one computer chip, and each year better software shrinks the fraction. Even if the rental price of chips doubled or tripled because of scarcity, the cost of replacing an hour of a person's mental work would still be far below that person's cost of living, and falling. For physical work the objection has more force, because robots have to be manufactured in the hundreds of millions before they are plentiful, and that will take longer. It is one reason this guide expects physical wages to hold up longer than mental ones.

There is a further limit even in the long run. A machine economy that keeps growing eventually presses against what cannot be manufactured: land, sunlight, minerals. Economists since Ricardo have understood what happens to a population that can multiply without limit on a fixed base of land. Its members end up earning their own cost of reproduction, and the surplus goes to whoever owns the land. Machines are such a population. In the long run their work is priced at what it costs to build and run them, and so is any human work that competes with theirs.

WHERE THE CHAIN STANDS

- A machine substitute caps the wage at the price of machine work (**P7**), and that price tends toward the cost of building and running machines (**P8**).
- While necessary tasks remain that only people can do, those people gain; when none remain, the gain reverses (**P9**). Average wages may therefore rise during the transition and are a poor warning signal.
- Machine cost is below the cost of supporting a person (**P10**).
- Under the premise, therefore, the wage stops carrying the economy's output to most people (**P11**), and very probably stops covering the cost of living (**P12**), although output is higher than ever.
- The economy has not become poorer. One of the three channels of income (**D1**) has closed. The next two chapters ask what happens to production, and then who receives it.

CHAPTER 6

What happens to production

The last chapter ended with a bleak statement about wages. This one is its counterpart, and the two have to be held together. The same premise that takes away the value of work also describes the most productive economy that has ever existed. People who ask whether the economy will "cease to function" after labor is automated usually have one of three different things in mind:

1. Will it still be able to *produce*?
2. Will producers still be able to *sell* what they make?
3. Will it still *provide* for the population?

This chapter answers the first question. Chapter 7 answers the other two.

Output rises

> **PROPOSITION 13 Automation raises what the economy can produce**
> A producer replaces a person with a machine only when the machine does the task at lower cost, which means using up fewer resources for the same result. The resources saved are available to produce something else. Every act of automation therefore raises the total that the economy is able to produce.
> Rests on: A4, A2

Nothing in the premise destroys a factory, a farm or a power station. Every good that could be made before can still be made, with fewer inputs. Whatever problems the post-labor economy has, a shortage of goods is not among them. This sounds too obvious to need saying, but a great deal of writing about automation slides from "people lose their wages" to "the economy collapses," as if the two were the same event. They are not. The first is about distribution, and the second would be about production.

Why growth has had a speed limit

To see how much more the economy could produce, we need to know what has limited its growth so far.

Output comes from combining work, tools, nature and knowledge (A2). Suppose you try to grow an economy just by building more tools. For a while it works. But the number of workers is fixed, and the tools pile up on them. The first laptop makes an office worker far more productive. The second adds a little. The tenth adds nothing. Economists call this diminishing returns, and it means that an economy cannot grow for long by adding tools to a fixed workforce.

Because of this, lasting growth in output per person has come almost entirely from the fourth input, knowledge: better recipes for combining the other three. In the countries at the technological frontier, it has run at about 2 percent a year for a century and a half, remarkably steadily. And new knowledge is itself produced by work, the work of researchers and engineers, whose number is limited by population and schooling.

So human work has been the speed limit twice over. It limits how much use can be made of additional tools, and it limits how fast new knowledge arrives.

> **PROPOSITION 14 Machine labor removes the population limit on growth**
> If machines can do the work, then work becomes something that can be manufactured, like any other tool. Tools and "workers" can then be multiplied together, and adding more of both does not run into diminishing returns in the way that adding tools to a fixed workforce does. The growth of the economy is then limited by how fast it can turn its output into more machines, and by whatever inputs still cannot be manufactured.
> Rests on: D3, P13

There is a second effect on top of the first. If machines can do research, then the production of knowledge also stops being limited by the number of human researchers.

How fast?

How much faster the economy would grow is the most disputed question in this whole field, and honest people who have studied it disagree by an order of magnitude. I will set out both sides and then say what the argument of this guide needs.

The case for explosive growth. In the standard models of economic growth, once the role of work can be filled by something that accumulates, growth speeds up without any further assumption. Output is reinvested in machines, the machines produce more output, and the loop has no fixed factor to slow it. Several careful studies conclude that growth rates of 20 or 30 percent a year, which would double the economy every three years or so, are plausible once nearly all tasks are automated. There is precedent for a change of this kind. Before the Industrial Revolution the world economy doubled about once in a thousand years, and since then it has doubled about once every twenty.

The case for bottlenecks. An economy is a chain, and a chain moves at the speed of its slowest link (A6 again). Even if most tasks are done a thousand times faster, the things that take time will still take time: pouring concrete, growing crops, running a clinical trial for years, getting a permit, waiting for a court. Things that cannot be manufactured will limit what can be built: energy first of all, then copper, land and water. And people will limit the pace through regulation and through simple slowness to adopt. On this view AI adds a percentage point or two to growth for some decades, as electricity and computers did.

Both sides are describing something real, and they are closer than they sound once you ask what the bottlenecks *are*. The explosive case depends on machines doing physical work at scale, because the bottleneck camp is right that an economy made of atoms cannot be sped up by thought alone. A world with machine minds and human hands has a hard limit on growth, and the limit is the hands. A world with both machine minds and machine hands does not have that limit, but it has the limits of energy and materials, and of the years it takes to build the factories that build the machines.

For the argument of this guide, the exact number does not matter, and I will lean on it as little as I can. What matters are two weaker statements that both sides accept. First, under the premise the economy's capacity to produce is much greater than today's and grows at least as fast.

Second, the faster it grows, the faster machine labor becomes plentiful, and so the faster the wage results of Chapter 5 arrive. Fast growth is good news about the size of the pie, and it shortens the time available to settle who gets it.

Where the scarcity goes

When one input becomes plentiful, the income it used to earn does not disappear. By **P1** it must be paid to someone. By **P2** it is paid to whatever is still scarce.

> **PROPOSITION 15 Income moves to what cannot be multiplied**
> As machine labor becomes plentiful, the share of income paid for work falls, and the same income is paid to the owners of whatever remains scarce. While machines are being built out, that includes the machines themselves and the things needed to make and run them: chips, power, the sites where both can be had. In the long run, machines are reproducible and earn only an ordinary return, and the surplus goes to what cannot be reproduced: land and location, energy and raw materials, legal permissions, and market positions protected from competition.
> Rests on: **P1**, **P2**, **P8**

This is an old result in a new setting. Ricardo worked it out for an economy of farmers two hundred years ago. As the population grew on a fixed amount of land, he argued, the wage would stay near subsistence, the return on tools would be competed down, and the surplus would go to the landlords as rent. He was wrong about his own century, because people did not multiply like his model said and technology outran them. The reasoning was sound, and it applies exactly to a population of machines that *can* multiply without limit.

Two practical consequences are worth drawing out, because they cut against a common picture.

The common picture is that "the owners of the AI" take everything. That may happen, but it does not follow from the economics. If many firms can build capable AI, and it is easy for a customer to switch between them, then competition pushes the price of machine work down to cost (**P8**) and the makers of AI earn ordinary profits, as the makers of personal computers eventually did. In that case the gains go first to the customers, as low prices, and then, as the economy expands, to the owners of the things the expansion runs short of. The evidence of 2026 is mixed on which way this is going, and Chapter 11 looks at it.

The second consequence concerns what a society could tax or share. Land, energy sites, mineral rights, radio spectrum and legal permissions have a property that economists have prized since Henry George: they stay where they are. A tax on the rent of something that cannot be moved or hidden, and that nobody made, does not reduce the supply of it. The post-labor economy concentrates income in exactly the kind of asset that is easiest to tax without doing harm. Chapter 10 comes back to this.

What a faster-growing, labor-free economy does to interest and asset prices

One more piece of machinery needs describing, briefly, because it shapes the transition.

When there are very profitable things to build, the demand for the funds to build them is high, and the price of funds, the real interest rate, is pulled up. Under the premise there is an almost unlimited amount of profitable building to do: every machine that can replace a wage at a

fraction of its cost is a profitable investment. Theory therefore predicts high real interest rates during the build-out, perhaps much higher than any in living memory.

High rates lower the present value of every fixed stream of future payments. At the same time, incomes shift from wages to profits and rents. So the build-out revalues assets in a predictable direction. Claims on scarce inputs to the machine economy gain. Claims that depend on wages being paid lose: consumer loans, mortgages in cities that live on office work, shops that serve the middle of the market, and the tax base of any government that leans on payrolls. None of this is advice about any particular asset, and timing such changes is notoriously hard. The point is structural: a great deal of the existing financial system is a set of promises that assume wages will continue, and Chapter 7 shows why that makes the transition fragile.

WHERE THE CHAIN STANDS

- Under the premise the economy can produce more than ever (**P13**), and the population limit on its growth is gone (**P14**). How much faster it grows is disputed, and depends above all on machines doing physical work.
- The income that wages used to carry moves to whatever stays scarce: at first the machines and their inputs, in the end the things that cannot be manufactured (**P15**).
- So the first of the three questions has a clear answer. The economy does not cease to function as a system of production. The open questions are whether producers can sell what they make, and whether the population can buy it.

CHAPTER 7

Who receives the output

We now have an economy that can produce more than ever (Chapter 6) and in which work no longer pays (Chapter 5). This chapter asks where the output goes. It also takes up the oldest objection to the whole line of thought, the one most people raise first: *if nobody earns a wage, who will buy the products?*

The default: income follows ownership

Recall that income can come only from work, from ownership or from transfers (D1). Remove the first and two remain.

> **PROPOSITION 16** **Without wages, income follows ownership**
> Under the premise, all of the economy's income is paid to the owners of machines, land, natural resources and legal rights. Any person's claim on output is then the return on what they own plus whatever is transferred to them.
> Rests on: D1, P1, P11

So the question "who receives the output?" becomes "who owns things?" The answer for the United States is on record.

Table 1. Who owns what in the United States. Wealth is far more concentrated than income, and ownership of companies is more concentrated than wealth as a whole. PLACEHOLDER VALUES.

Group of households	Share of all wealth	Share of company shares	Share of income today
Wealthiest 1%	31%	50%	—
Next 9%	36%	37%	—
Middle 40% (50th to 90th percentile)	30%	12%	—
Less wealthy half	2.5%	1%	—

Income today is unequal, but it is much less unequal than wealth, and the reason is wages. Most people own little (A5) and nonetheless receive a real share of output, because they sell work. Take the wage away and the distribution of income moves toward the distribution of ownership.

PROPOSITION 17 **The default distribution**

If nothing else changes, the distribution of income in a post-labor economy converges on the distribution of wealth, which is several times more concentrated than the distribution of income is today. The less wealthy half of American households, who now receive a meaningful share of national income, own about a fortieth of national wealth.

Rests on: P16, A5

The position of a typical household is worse than its share of wealth suggests, for two reasons. First, most of what a middle household owns is its house and its retirement accounts. A house provides shelter but pays no cash. Second, most households carry debts that were taken on in the expectation of wages: a mortgage, a car loan, student loans. A household with a home worth $400,000, a mortgage of $250,000 and a retirement account of $100,000 is comfortably middle class today. With no wage it has payments to make and a few thousand dollars a year of investment income to make them with.

That is the default. It is what the rules of a market economy produce if they are simply left running under the premise. Whether they will be left running is the subject of Part III.

"But who will buy the products?"

Here is the objection in its strongest form.

OBJECTION **An economy without wage earners has no customers**

Workers are also consumers. If machines replace workers, the workers have no income; if they have no income, they cannot buy what the machines make; if nobody buys, the firms that own the machines have no revenue and the machines have no value. So mass automation is self-defeating. Either it will stop short, or firms will have to keep paying people, or the whole system will seize up.

There is a story, probably invented, that a Ford manager once showed the union leader Walter Reuther a line of new automatic machines and asked how he planned to collect union dues from them, and that Reuther asked how Ford planned to get them to buy cars. The exchange is quoted because it feels unanswerable. It has an answer, and the answer is in P1.

Every dollar of output is somebody's income. That is true with or without workers. If a factory sells $1 billion of goods and pays no wages, the $1 billion does not vanish. It is paid to the factory's suppliers and owners. *They* now have the income that the workers used to have, and they can spend it. The firm has lost its workers as customers and gained its owners as customers, for the same total. A firm does not care whose dollars it receives.

"But the owners are few, and a few people cannot eat a million loaves of bread." True, and so the economy stops making a million loaves of bread for them. It makes what the people with the income want. There are two things they can do with it.

They can *consume*: larger houses, private aircraft, staff, art, medical care at the frontier, security, travel, political influence. Wants are unlimited (A1), for the rich as for everyone.

Or they can *invest*: use the output to build more machines, more power plants, more data centers. Under the premise this is very attractive, because the return on new machines is

high (Chapter 6). An economy can direct a large part of its output to building its own future capacity for a long time. China has invested more than 40 percent of its output every year for two decades.

> **PROPOSITION 18 There is no demand paradox, only a change in what is produced**
> A post-labor economy does not run out of buyers. Total income still equals total output. What changes is whose wants direct production. Spending shifts from what wage earners buy toward what the owners of capital and land want to consume, toward investment in more machines, and toward whatever governments purchase.
> Rests on: P1, A1

History has many economies organized this way, in which most of the people were not the customers production was for. In the slave plantations of the Caribbean and the American South, output went to export and to the planters. In several oil states of the Persian Gulf today, the economy runs on a resource that needs few local workers, and the population lives on what the state passes on. Pharaohs directed the surplus of the Nile into monuments. Such economies can last for centuries. They show that "the circular flow must include the masses" is not a law of economics. It is a feature of economies in which the work of the masses is needed.

What about Henry Ford, who famously doubled his workers' pay to five dollars a day in 1914, supposedly so that they could afford his cars? He did it to stop his workers quitting. No firm can create its own customers by paying wages, because its workers spend only a tiny part of their pay on its products. Each firm gains by cutting its own wage bill and loses only when all the others cut theirs. So even if every firm understood that mass purchasing power served them all, no single firm would have a reason to preserve it. The purchasing power of the public is a shared resource that no firm owns, and markets do not look after resources that nobody owns.

So the answer to the second of our three questions is yes: producers can sell what they make. They sell to different buyers. A good way to keep this in mind is that the post-labor economy does not have a demand problem. It has a *whose-demand* problem.

The transition is where demand can break

That was the destination. The journey is more dangerous, and here the objection has real force. The economy can move from one pattern of spending to another, but it cannot do so quickly without things breaking, for three reasons.

Wage earners spend what they get; owners mostly do not. A household living on wages spends nearly all of its income. A wealthy household that receives an extra dollar of profit spends a small part and saves the rest. When income moves quickly from wages to profits, total spending falls unless investment or government spending rises at the same moment by the same amount. There is no mechanism that guarantees the timing.

Debts are fixed in dollars. American households owe about $18 trillion, most of it mortgages, on the assumption that wages will arrive. When wages stop, payments stop. Defaults damage banks, banks lend less, house prices fall, and households who still have jobs cut spending because they feel poorer. This is the mechanism that turned the 1929 crash into the Great Depression and the 2007 housing bust into the Great Recession.

Recessions are when the cuts happen. Firms are reluctant to dismiss workers in good times and quick to do it in bad times. Studies of recent decades find that routine jobs were destroyed almost entirely during recessions and did not come back in the recoveries. A downturn caused by displacement therefore causes more displacement.

> **PROPOSITION 19 A fast transition can produce a slump in the middle of abundance**
> If wage income falls faster than other spending rises to replace it, the result is an ordinary recession on top of the structural change: falling sales, defaults, failing lenders and further layoffs. The economy's ability to produce is unharmed throughout. What fails is the flow of spending.
> Rests on: **P11**, **P18**

Two things can offset the fall in wage-earners' spending while it happens. One is a boom in investment: if firms are spending hundreds of billions of dollars a year on data centers, chips and power plants, that spending is income for someone and can hold total demand up even while employment weakens. The result is growth without jobs. The other is the government, which can put money directly into households' hands and borrow or tax to pay for it, as the United States did on a very large scale in 2020. Part IV argues that the United States will see both, in that order.

What gets cheap, and what does not

So far this chapter has been about incomes. The other half of anyone's standard of living is prices, and here the news is good, with an important exception.

> **PROPOSITION 20 Machine-made things get cheap; scarce things do not**
> Competition pushes the price of anything machines can make toward the cost of making it, and that cost falls with the cost of machine labor. Anything whose supply is limited by nature, by position or by law does not get cheaper, and tends to get more expensive relative to everything else.
> Rests on: **P8**, **P15**

On the cheap side: manufactured goods of every kind; software and entertainment; transport, once vehicles drive themselves; and, most strikingly, expertise. A consultation with a doctor, a lawyer, an accountant or a tutor is expensive today because it uses the time of a highly trained person. Under the premise the knowledge is available at the cost of electricity. In time the same applies to building things and growing food, as machines take over the physical work.

On the other side: land in places people want to live, which is most of the cost of housing in every prosperous city. Energy, at least while the build-out strains supply. Anything rationed by rank, such as a place at a famous university or a house in the best school district, since rank cannot be mass-produced. Anything the law requires a licensed person to provide. And the time and attention of other human beings.

This split has a consequence that is easy to get wrong in both directions. It is *not* true that falling prices make the loss of wages harmless, because a person still needs somewhere to live, and land is exactly what does not get cheap. But it *is* true that the income a person needs for a

materially decent life falls a long way, perhaps to a fraction of today's median wage, provided they have a place to live. The post-labor economy makes the cost of providing for everyone lower than it has ever been. Whether everyone is provided for is a separate matter, and it is not an economic one.

The three questions, answered

We can now answer the question from the start of Chapter 6: does the economy cease to function?

- **As a system of production**, no. It produces more than ever (P13).
- **As a circuit of buying and selling**, no, once it has adjusted (P18). During the adjustment it may well break down for a time, as economies have before (P19).
- **As a system that provides for the population**, yes, by default (P17). The channel through which most people received their share has closed, and nothing in the market reopens it.

The economy keeps functioning. What it stops doing is functioning *for* most people, unless something other than the labor market connects them to it.

WHERE THE CHAIN STANDS

- Without wages, income is the return on what a person owns plus what they are given (P16). Ownership in the United States is far more concentrated than income (P17).
- "Who will buy?" has an answer: whoever receives the income. Production turns toward the wants of owners and toward investment (P18). No firm has a reason to preserve the purchasing power of the public.
- A rapid shift can still cause a slump, through lower spending and unpayable debts (P19).
- Machine-made goods and services become cheap, so providing for everyone becomes cheap, but land and other scarce things do not (P20).
- What remains is to ask what people can still be paid for (Chapter 8), and then who has the power to decide the distribution (Part III).

CHAPTER 8

What people can still be paid for

The wage ceiling of Chapter 5 applies wherever a machine is a substitute for a person. There is one kind of work to which it does not apply even under the full premise, and this chapter measures how much that exception is worth.

Where the buyer wants a person

> DEFINITION 5 **Human-required work**
> Work is human-required when the buyer's purpose is not served by a machine however capable the machine is, either because a human being is part of what the buyer wants, or because a rule demands that a human being do it.

It helps to sort this work by the reason a person is required.

- **Presence.** Some services are valued because another person is there: much of the care of small children and the old, the work of clergy, coaches and mentors, hospitality at its best, companionship. A machine may do every task involved and still not be what was wanted.
- **Contest and performance.** People want to watch people. Computers have played chess better than any human since 1997, and far more people now follow human chess than did then; almost nobody watches computers play each other. The same holds for sport, live music and theater.
- **Provenance.** Some goods are valued for where they came from: a handmade table, a painting by a known artist, a letter someone wrote themselves. The machine-made version may be better in every measurable respect and still be a different product.
- **Accountability.** Some roles exist so that a person answers for a decision: the judge, the juror, the elected official, the company director, the engineer who signs the drawing. We want someone who can be blamed, sued, dismissed or voted out.
- **Legal reservation.** The law reserves some work for licensed people: prescribing drugs, arguing in court, flying passengers. This lasts as long as the law does.
- **Status.** Human service can be bought as a display of wealth precisely because it is unnecessary: the butler, the tailor, the concierge. The economist Thorstein Veblen described the pattern in 1899.

These are real, and under the premise they are all that is left. Wages will not be zero. The question is whether this work can do what the labor market does today, which is to give most households most of their income.

The arithmetic of human-required work

Under the premise, the only wages paid are for human-required work. That gives a simple identity.

PROPOSITION 21 The wage share equals the human-required share of spending

In a post-labor economy, total wages equal total spending on human-required work. The share of national income paid as wages is therefore equal to the share of all spending that buyers choose, or are obliged by law, to direct to work done by people.

Rests on: **P1**, **P11**, **D5**

Today about 60 percent of American national income is paid for work. How much of all spending would people direct to a person when a machine could do the task as well and for less?

The evidence of the past fifty years is not encouraging. People said they valued the bank teller and then used the cash machine. They valued the travel agent, the shop assistant and the local newspaper, and then used the website. Wherever a machine has offered roughly the same result at a lower price, most buyers have taken it most of the time. The preference for a human is real, and it is strongest in a fairly narrow range of things, mostly in the six categories above. A generous guess is that it covers a tenth of spending. A less generous one is a few percent.

History offers a check. In societies where a small class held most of the income, the largest occupation was usually service to that class. In England and Wales in 1901, domestic service was the largest single category of employment, with around one and a half million indoor servants. That is what an economy does with people when the main source of demand for their work is the households of the rich. It employed them in large numbers and at low pay, because many competed for the places.

PROPOSITION 22 Human-required work reflects the distribution of income and does not correct it

The pay for human-required work is set by how much those with income choose to spend on it, divided among everyone who seeks such work. If income is concentrated, many people compete to serve few, and pay is low for all but the most sought-after. If income is widely spread, people buy human services from one another, and pay can be respectable. In both cases the distribution of non-wage income comes first, and the wages follow from it.

Rests on: **P21**, **P16**

These are two very different societies, and it is worth picturing each.

In the first, the owners of the machine economy are the only people with much to spend. Around them grows an economy of attendants: household staff, private tutors, security, entertainers, craftspeople, companions. A few performers and personalities with a mass audience become very rich, since one star can serve millions. Everyone else competes for the remaining places.

In the second, every household has an income from some other source, and people spend part of it on each other. One person coaches the neighborhood children at football and pays another to teach her the cello. There are more musicians, more small restaurants where the owner cooks, more teachers with five pupils. None of this is necessary in the sense that machines could not do it. People do it because they value doing it and value having it done by someone they know. This is a real economy, with prices and earnings, sitting on top of the machine economy, and it could be a good one. But notice that it exists only *because* incomes were already spread. It is a result of the answer to the distribution question and cannot be the answer.

A twist: scarce things attract spending

There is one argument for a larger role for human work in the long run, and it deserves a hearing, because it comes from the same principle as everything else in this guide.

As machine-made things become very cheap, people do not necessarily spend less on them in total, but past experience says the share of spending going to cheap, plentiful things tends to fall and the share going to scarce things tends to rise. This is Baumol's pattern again. In a society ten times richer than ours, where every manufactured thing costs almost nothing, a large part of spending might go to the two things that remain scarce: land and people.

The arithmetic is worth doing. Suppose the American economy grew to ten times its present size, and just 5 percent of spending went to human-required work. That 5 percent would be about half of today's entire national wage bill. Divided among a workforce the size of today's, it would pay roughly half of today's average wage, in a world where most things cost far less.

So in the far future of a very rich post-labor economy, human-required work could amount to something substantial, especially if incomes are broadly spread. Two cautions apply. It depends on the economy first growing many times over, which lies beyond the ten-year horizon of Part IV. And it is still governed by **P22**: the spending has to come from people who already have incomes.

Owning and wanting

One last candidate for "what people still do" is often put forward: people will be the entrepreneurs and the directors, deciding what the machines should make.

Under the full premise this does not survive as work. Running a company, spotting an opportunity and allocating capital are tasks, and the premise is that machines can do tasks. What remains, after the tasks are subtracted, is a position. An owner is the one on whose behalf the machines act and to whom the proceeds are paid. A customer is the one whose wants production serves. A citizen is the one whose voice counts in setting the rules. These are roles a person holds because of what they own, what income they have and what rights they hold. They are the three ways in which people differ from horses, listed at the end of Chapter 3, and they are positions of entitlement. They cannot be earned by effort after the fact. Whether most people hold them, and how securely, is the question the rest of this guide turns on.

A note on purpose

This guide is about economics, and I will not pretend to settle what people would do with lives that did not need to be spent working. But one piece of evidence bears on the forecasts that follow.

People who lose their jobs become much less happy, by more than the loss of income explains, and they do not fully adapt. People who retire on a secure pension do not show this effect, and neither, as far as anyone can tell, did the leisured classes of earlier centuries. The difference seems to lie in whether the absence of work is chosen, respected and financially secure, or imposed, stigmatized and precarious. That suggests the damage is done less by the absence of work than by the terms on which it is absent. It also suggests, as a matter of political prediction, that societies will fight hard to keep "jobs" in existence long after the work is needed, because a job is how people currently receive not only income but standing. This will matter in Part IV.

WHERE THE CHAIN STANDS

- Under the premise, wages are paid only where a person is wanted as a person or required by law (**D5**).
- The wage share of income then equals the share of spending people choose to direct to such work (**P21**), which on present evidence is a small fraction of today's wage share.
- This work mirrors the distribution of other income (**P22**). With concentrated income it is an economy of servants and stars. With broad income it is an economy of people doing things for one another.
- So Part II ends with a problem and no solution inside the market. Production is abundant; the channel that connected most people to it is closed; what is left of paid work depends on a distribution of income that has to come from somewhere else. Where it comes from is a question of power.

PART III

Who decides

The market does not settle who receives the output of a post-labor economy. Power does. These chapters ask where the majority's power has come from, what happens to it, and what could be done with it.

CHAPTER 9

Who decides

Part II ended with an economy that produces abundantly and a majority with no claim on the product. The market does not supply the missing claim. So the outcome depends on a decision, and this chapter asks who is in a position to make it.

At the end of Chapter 3 I listed three ways in which a population of people differs from a population of horses: people are customers, owners and citizens. Two of the three have now been examined and found wanting. Being a customer gives no protection, because production follows income, wherever the income is (P18). Being an owner protects those who own, and most people own little (P17). That leaves the third. People govern themselves, and governments can tax and transfer. What is that worth?

Why majorities have had a share

Start with an uncomfortable axiom.

> **AXIOM 8** **Shares follow leverage**
> Over the long run, a group's share of what a society produces reflects what that group is able to withhold from others or impose on them. Entitlements that are not backed by some form of leverage tend to erode.

This is a claim about how societies have in fact behaved, and the reader should want evidence for it. There is a good deal.

Wars that needed everyone. The great expansions of voting rights, progressive taxation and social insurance in the rich countries cluster around the two world wars, when states needed the mass of the population as soldiers and as workers in war industries. Top income tax rates rose sharply in the countries that mobilized mass armies and much less in those that did not. Britain widened the vote in 1918 and built its welfare state after 1945. The United States passed the GI Bill for sixteen million veterans.

Workers who could stop production. Bismarck created the first national social insurance in the 1880s, by his own account to draw workers away from the socialists. The American New Deal followed waves of strikes. The golden age of rising wages after 1945 was also the high-water mark of union membership. In each case, concessions were made to people who could shut things down.

Plagues that made labor scarce. After the Black Death killed a third or more of Europe's population, the wages of the survivors rose for generations despite laws passed to hold them down. The historian Walter Scheidel, surveying inequality across several thousand years, found that it has fallen substantially only after mass-mobilization wars, revolutions, state collapses and catastrophic plagues: events that either made ordinary people's labor scarce or put weapons in their hands.

States that do not need their people's work. The clearest evidence is the reverse case. Where a government draws its revenue from oil or minerals and not from taxing the work of its citizens, it has less need of their consent, and the long-standing finding of political scientists is that such states are on average less democratic and less accountable. The slogan of the American Revolution runs in both directions: no taxation without representation, and little representation without taxation.

So the working majority of the industrial countries has had three sources of leverage. They could withhold their work, which is the strike. The state needed them to fight and to pay taxes, which made governments attentive. And they could vote, with the implied possibility of disorder if votes were ignored.

What automation does to leverage

> **PROPOSITION 23 Automation removes the leverage of being needed**
> Under the premise, the first source of leverage disappears by definition: work that is not needed cannot be withheld. The second weakens as armies and police forces automate and as the state's revenue stops coming from wages. What remains to the majority is the third: its formal political rights, and the willingness of institutions to honor them.
> Rests on: A8, P11

A group of writers has named this prospect "the intelligence curse," by analogy with the resource curse of the oil states: if a country's income flows from machines and not from its people's work, those who control the state have less reason to invest in the people or to listen to them.

The point is easy to overstate, and precision matters here. Formal rights are powerful. Courts, constitutions, elections and a free press are real constraints, and in the countries that have them they have survived severe tests. The claim is narrower. Until now, wherever democracy has existed, it has existed alongside an economy that needed the majority's work. We have no example of a democracy lasting for generations in which most citizens were economically unnecessary. The oil states are the nearest examples of unnecessary majorities, and most of them are not democracies. Nobody knows whether formal rights alone can hold a fair distribution in place indefinitely. They have never had to.

There is also an encouraging pattern in the same evidence. Where strong institutions were in place *before* the windfall arrived, the windfall was shared. Norway found oil as an established democracy and put the proceeds into a public fund that is now worth more than $300,000 per citizen. Alaska wrote a dividend from its oil fund into its constitution and laws in the years around 1980, and every resident has received a payment each year since. Nigeria, Angola and Venezuela found oil with weaker institutions, and the proceeds were captured. The lesson is about order. Institutions that are settled before the money flows tend to keep the money flowing to the public. Institutions that are built afterward have to be extracted from those who already hold it.

PROPOSITION 24 **The window**

The majority's leverage is greatest before automation is complete, while its work is still needed, its taxes still fund the state and its numbers still fill the army. It declines as automation proceeds. Whatever lasting claim the majority is to have on the output of the machine economy therefore has to be established during the transition. The period in which the claim is most needed is the period in which it is hardest to win.

Rests on: A8, P23, P9

P9 adds a complication. During the transition, average wages may be rising and statistics may look healthy, which is when the case for acting seems weakest. By the time the need is obvious, part of the leverage is gone.

The state has its own problem

It would be a mistake to treat the government as an umpire standing outside the game. It is a player with a budget, and the premise wrecks that budget.

PROPOSITION 25 **The state must find a new tax base**

The United States government draws about five sixths of its revenue from taxes that fall mainly on work: the income tax on individuals and the payroll taxes that fund Social Security and Medicare. As the wage share of income falls, that revenue falls with it, at the same time as the demands on public support rise. The state must then tax capital, land, consumption or machine labor directly, or own income-producing assets itself, or borrow and create money without limit.

Rests on: P11, P3

This cuts in two directions, and both matter for the forecasts.

It means the transition will produce a fiscal crisis whether or not politicians wish to address inequality. Falling payroll-tax receipts will threaten Social Security and Medicare, the two most popular programs in American government, on which the most reliable voters depend. The state will be forced to go where the money is, because it will have no alternative. So there will be taxes on the machine economy. The open question is what is done with the proceeds.

It also means the state changes its nature. A government that lives on taxes paid by a hundred and sixty million workers has to care, at least a little, whether those workers prosper. A government that lives on taxes paid by a few hundred firms that own the machines is a different kind of thing. It has every reason to care whether those firms prosper, and its concern for everyone else rests on the vote alone. This is the mechanism behind the resource curse, and it does not require anyone to be wicked.

There is one more fact, and it is favorable. The machine economy is unusually easy to tax. A great deal of twenty-first-century wealth has been hard for governments to reach because it is intangible and mobile: a patent can be moved to Ireland with a signature. The capital of the machine economy is the opposite. It consists of data centers the size of towns, power plants, chip factories and fleets of robots. It sits on land, draws power from a grid, needs permits, and can be seen from orbit. It also depends on the state at every turn, for land, power, security and law. A government that wants to tax it or take a share of it can.

Sharing is affordable

Suppose the political will existed. Could a post-labor economy afford to provide for everyone? The arithmetic is short.

There are about 265 million adults in the United States. To pay every one of them $1,000 a month would cost about $3.2 trillion a year. That is roughly a tenth of the present output of the economy, and less than the federal government already collects in taxes on work. Part of it would replace existing programs. If the economy doubled in size, as even cautious forecasts of the premise allow, the same payment would take a twentieth of output. And **P20** says that what the payment buys would grow over time, because the things machines make get cheaper.

> **PROPOSITION 26** **Providing for everyone takes a small and shrinking share of output**
> In a post-labor economy, an income sufficient for material comfort can be given to every person at a cost that is a modest fraction of total output and that falls as the economy grows. The obstacle to providing for everyone is not a shortage of resources.
> Rests on: **P13**, **P20**

This is the most hopeful result in the guide, and it should be given its full weight. In every earlier society, lifting everyone out of want would have required the better-off to give up a great deal, because there was not enough to go round. In the post-labor economy the owners of the machines could provide for everyone else out of a small part of the *growth* in their income, and still become richer every year than anyone has ever been. Self-interest alone gives them reasons to do it: a population with nothing is a threat to property, and a tenth of output is a modest price for peace. Historically, though, elites have often failed to make cheap concessions in time, and the cheapness of sharing does not guarantee that it happens.

Who owns the machine economy?

A final fact shapes the politics. “The owners” are not five men in California.

Ownership of the leading AI companies and of the firms that supply them is held through the stock market and private funds, and so belongs mostly to the same households that own everything else: about half to the wealthiest 1 percent and most of the rest to the next 9 percent (Table 1). That upper tenth is some thirteen million households. They vote and donate at much higher rates than everyone else. They are the default beneficiaries of the premise.

But many of them are also among its first casualties. The upper tenth of American households is full of lawyers, doctors, engineers, managers, consultants and financial professionals, people sometimes called the working rich, whose wealth is real and whose current income is mostly a salary. Chapter 12 will show that mental work is automated before physical work. So the first large group to lose its earning power will be the most educated, articulate, organized and politically connected class in the country, and its members will have enough assets to feel what they are losing and too few to live on.

That is a very different political event from the decline of factory towns after 2000, which fell on people with little voice and was ignored for fifteen years. It is the main reason I expect the political response to automation to come faster, and to be more forceful, than the response to earlier shocks.

WHERE THE CHAIN STANDS

- Majorities have had a share of output because they were needed, as workers, soldiers and taxpayers (A8). Automation removes that need (P23), leaving formal political rights to do work they have never had to do alone.
- Leverage is greatest early. Lasting claims have to be established during the transition (P24), when average statistics may still look fine.
- The state will be forced to tax the machine economy to survive (P25), and the machine economy is physically easy to tax. What it does with the proceeds is not forced.
- Providing for everyone would take a small share of output (P26). The constraint is political.
- The first to lose their earnings will be the professional class, which has unusual political weight.
- The next chapter sets out the choices.

CHAPTER 10

The choices

The argument so far has produced a clear specification of the problem. Production is abundant (P13). The wage, which connected most people to it, has stopped doing so (P11). The market supplies no replacement (P17, P22). A replacement is affordable (P26), and it has to be put in place while the majority still has leverage (P24).

This chapter lays out what the replacement could be. It is organized by the logic of D1, which said that income can come only from work, from ownership or from transfers. Every proposal ever made for a post-labor economy does one of four things: it keeps the work channel open by rule, it opens the transfer channel, it opens the ownership channel, or it reduces how much income a person needs. I take them in that order, and then ask what the chain of reasoning says about choosing among them.

Keeping the work channel open by rule

The first family of responses tries to preserve the link between work and income by making sure there are jobs for people, whether or not machines could do them.

Reserving work for people. A law can require that a human being drive the truck, sign the diagnosis, teach the class or approve the dismissal. A union contract can forbid a kind of machine. Such rules exist and are spreading. The East Coast dockworkers won a contract in 2025 that bars fully automated equipment from their ports. Eight American states passed laws in 2026 requiring a human to review adverse decisions by health insurers. Bills requiring a human operator in heavy autonomous trucks have been introduced in many states, though governors have so far vetoed the ones that passed [10].

Reserving work has one great strength and three weaknesses. The strength is that it buys time, and by P24 time is what the majority most needs. The weaknesses are these. It costs output, because a task is done the expensive way. It leaks, because a firm, a state or a country that does not impose the rule can undercut one that does; it therefore holds best in work that is local and already regulated, such as health care, schools, courts and government, and worst in anything traded. And under the full premise it changes its nature. A job that exists only because the law forbids a cheaper machine is a transfer from customers to the job-holder, with the size of the transfer set by who happened to hold the protected position. Notice the pattern in who has won such protection so far: the dockworkers, who can close a port, won a ban, and unions without a chokepoint won the right to be notified.

Subsidizing and sharing work. Governments can pay part of a wage, insure workers against a fall in pay, pay for retraining, or shorten the standard week so that the remaining work is spread over more people. These fit the period of partial automation well. Shorter hours are how much of the last century's productivity gain was taken: the standard American week fell from about sixty hours to forty between 1900 and 1940. Retraining has a weaker record than its popularity suggests, and it has a logical limit, since it helps only if there is something to retrain

for (P5). All of these tools assume that human work is still worth something close to its wage. They stretch the transition and do not survive its end.

Guaranteeing work. The state can offer a job to anyone who wants one, in care, in public works, in community service. Under the full premise these are human-required jobs created by decree (D5), which makes the program a transfer with a condition attached. That is not an argument against it. The condition is what many people want. In a large 2026 survey, Americans asked how government should respond to job losses from AI preferred support earned through national service to an unconditional payment by seven to one [11]. A guaranteed job supplies structure and standing as well as income (Chapter 8). Its difficulty is administrative: no country has run such a program for tens of millions of people.

Opening the transfer channel

The second family pays people directly: a universal basic income, a negative income tax that tops up low incomes, or broader unemployment and retirement benefits.

What the evidence shows. The best American experiment gave a thousand low-income adults $1,000 a month for three years. They worked a little less, by one to two hours a week, and about four in a hundred left the labor force. Stress and hunger fell at first, and there was no lasting improvement in measured health [12]. Alaska has paid every resident a yearly dividend since 1982 with no detectable effect on total employment [13]. Studies in Kenya, Iran, Finland and Germany found small or no reductions in work. Across the evidence, people who are given money mostly keep working, and the benefits are real and more modest than advocates claimed. But there is a gap in all of it that matters for this guide: every study gave cash to people who could still find work. Nobody has tested cash in an economy where the demand for work has collapsed.

What it costs. Table 2 gives the arithmetic for the United States.

Table 2. The gross cost of a universal payment in the United States, taking 270 million adults, output of $33 trillion and federal revenue of $5.4 trillion in fiscal 2026. Today much of the cost could be recovered by taxing the payment back from higher earners. In a post-labor economy there are few earners to recover it from, and the net cost approaches the gross cost [10].

Payment to every adult	Cost per year	Share of output	Share of federal revenue
$500 a month	$1.6 trillion	5%	30%
$1,000 a month	$3.2 trillion	10%	60%
About $16,000 a year (the poverty line for one person)	$4.3 trillion	13%	80%
$2,000 a month	$6.5 trillion	20%	120%

Where the money would come from. It cannot come from the present tax base, most of which is the wage income that is disappearing (P25). It has to come from the income of the machine economy. Here P1 is reassuring: the dollars that stop being paid as wages do not vanish. They become profits and rents, and a tax on those at the rate now applied to wages would raise about what the wage taxes did. The base is there. Taxing it is a matter of will and design.

On design, economic theory offers some guidance. A tax aimed specifically at robots or at AI usage is, at best, a temporary measure: the standard result is that it is justified while today's workers cannot retrain and should fall toward zero afterwards [14]. Taxes on AI alone also raise very little at present. A 3 percent tax on AI revenue would raise about $3 billion for every $100 billion of revenue, against a need measured in trillions [10]. The efficient things to tax are the ones **P15** identified: the rents of what cannot be multiplied, such as land, energy sites and legal monopolies, and the windfall gains on assets that were already in place. Taxing those does not reduce the supply of anything [15]. Broad taxes on capital income and on consumption can raise far more, and they carry a cost that is larger in a post-labor economy than today, because when growth runs on reinvested profits, a tax on profits slows growth directly [9].

The weakness of transfers is durability. A transfer is a decision that a legislature makes again every year. In 2021 the United States expanded its tax credit for children, and the share of children in poverty fell from 9.7 percent to 5.2 percent, the lowest on record. The expansion lapsed after one year, and child poverty went back above 12 percent [16]. Compare Social Security, which has survived ninety years of attempts to cut it. The difference is that Social Security is universal, and its recipients believe they have paid for it and own it.

Opening the ownership channel

The third family goes to the root. **P16** says that when wages are gone, income follows ownership. So change who owns.

There are several ways. A *public fund* can hold a share of the nation's productive assets and pay everyone a dividend, as Alaska's does with oil. *Personal accounts* can be funded for every citizen. *Employees* can be given shares in the firms they work for. Each of these has moved from theory to live politics in the United States with surprising speed.

- The 2025 tax law created investment accounts for children with a $1,000 federal deposit for each child born from 2025 to 2028, and by October 2026 the government had begun enrolling some sixty million children automatically.
- In June 2026 Senator Sanders introduced a bill to move half the shares of large AI companies into a public fund paying every resident about $1,000 a year.
- In the same month President Trump said the government might take stakes in the leading AI companies, and OpenAI was reported to have proposed giving it 5 percent.
- The federal government already holds about a tenth of Intel and stakes in a number of mining and chip companies.
- The chief executive of OpenAI proposed in 2021 a fund fed by a 2.5 percent annual levy, payable in shares, on the value of large companies and of land, which on his arithmetic would pay each adult about $13,500 a year within a decade [10], [17].

The case for ownership follows from three earlier results. It is the channel that still works under the full premise (**P16**). A share of capital income grows with the machine economy automatically, where a fixed payment has to be raised by a new vote each time. And it is the most durable form of claim, for a reason worth spelling out.

PROPOSITION 27 Claims that are property outlast claims that are appropriations

A claim on the machine economy's output is more durable the less it depends on a majority's continuing leverage. A dividend from assets held in the people's name, protected by the same law that protects all property, is more durable than a benefit defined in statute, which is more durable than an annual appropriation, which is more durable than a job preserved by a rule that needs constant enforcement.

Rests on: A8, P24

The reasoning is that property rights are defended by everyone who owns property, including the most powerful. To take away a citizen's dividend, a future government would have to do something that looks like expropriation, in a legal system the wealthy themselves rely on. To take away a benefit, it need only fail to renew it. Alaska shows the pattern: its dividend has survived forty years and several budget crises, and governors who cut it have paid for it at the polls.

Ownership has real difficulties too. The amounts needed are very large. To pay each of 270 million adults $12,000 a year from the returns on a fund, at a 4 percent payout, would take a fund of about $80 trillion, which is roughly half of all the wealth American households own today. A fund of that size cannot be bought. It could only be built by claiming a share of the *growth* in the value of the machine economy over many years, and the children's accounts enacted so far are smaller than this by a factor of several hundred. There is a problem of governance when the state is both a large owner of companies and their regulator. And a dividend requires profits, which most AI companies do not yet have.

Reducing the need for income

The fourth family works on the other side of the ledger. If the things people need are provided directly or made very cheap, a small income goes a long way.

P20 does much of this without any policy at all: under the premise, the price of anything machines make falls toward its cost. What policy can do is protect that process and extend it to the things that would otherwise stay expensive. Competition has to be kept alive among the suppliers of machine work, or the fall in costs is kept as profit and never reaches prices. Medical, legal and educational services become cheap only if the rules that reserve them for licensed people are changed to let machine expertise in. And housing becomes cheap only where building is permitted, because the cost of a home in a prosperous city is mostly the cost of the right to occupy land there.

This family has an appealing property: it does not require agreeing on who deserves what. It is also the one most exposed to the conflict described under the first family, since every rule that keeps a service expensive is also somebody's protected job.

What the argument says about choosing

No single one of these is sufficient, and any real response will combine them. The chain of reasoning does, however, rank them on the dimension that matters most.

Timing decides what is possible. By P24, leverage declines along the transition. Measures that create durable claims, above all ownership stakes and benefits that switch on automatically

when unemployment crosses a threshold, are worth far more if enacted early, because later they will have to be extracted from stronger hands by weaker ones. Measures that only buy time, such as reserving jobs, are worth having in order to enact the durable ones, and for little else.

The test of any proposal is one question. Does it give ordinary people a claim on the output of the machine economy that grows as that economy grows and that does not have to be won again every year? A proposal that passes this test addresses the problem this guide has derived. One that fails it is at best a way of managing the transition.

What is politically natural is not what is durable. The path of least resistance, in a country that prefers earned support and distrusts "basic income," runs through protected jobs, extended unemployment benefits and one-off checks. These are the least durable forms. The most durable form, a real ownership stake, is the largest departure from present arrangements. Part IV has to forecast which of these the United States will actually do, and it will not assume that the country does what the argument recommends.

WHERE THE CHAIN STANDS

- There are only four kinds of response: keep jobs by rule, pay transfers, spread ownership, or lower the cost of what people need.
- Job protection buys time at a cost and leaks through competition. Transfers are affordable and fragile. Ownership is durable and requires very large sums, which can only come from a share of future growth. Cheaper essentials help everyone and collide with protected jobs.
- Claims that are property outlast claims that depend on a yearly vote (**P27**), and the time to create them is early (**P24**).
- This completes the argument from first principles. Everything so far has been conditional on the premise. Part IV asks how much of the premise is true today, how fast the rest is arriving, and what the United States is likely to do about it.

PART IV

The next ten years

How much of the premise is true in November 2026, how fast the rest is arriving, six scenarios for the decade to 2036, and what I actually expect.

CHAPTER 11

Where things stand in November 2026

Everything in Parts II and III was conditional on the premise. This chapter is an inventory of the facts as they stand on the day of writing: what the machines can do, what they cost, how much money is behind them, what has happened to work so far, and what governments have done. The research notes in the repository give the source for every number [5], [6], [10], [18], [19].

The inventory has a clear shape, visible in Figure 3. The capability of the machines, and the money flowing to the companies that make them, are growing at rates with no precedent. The economy-wide statistics look ordinary. Any forecast has to account for both halves of that picture, and most public argument about AI consists of people pointing at one half and ignoring the other.

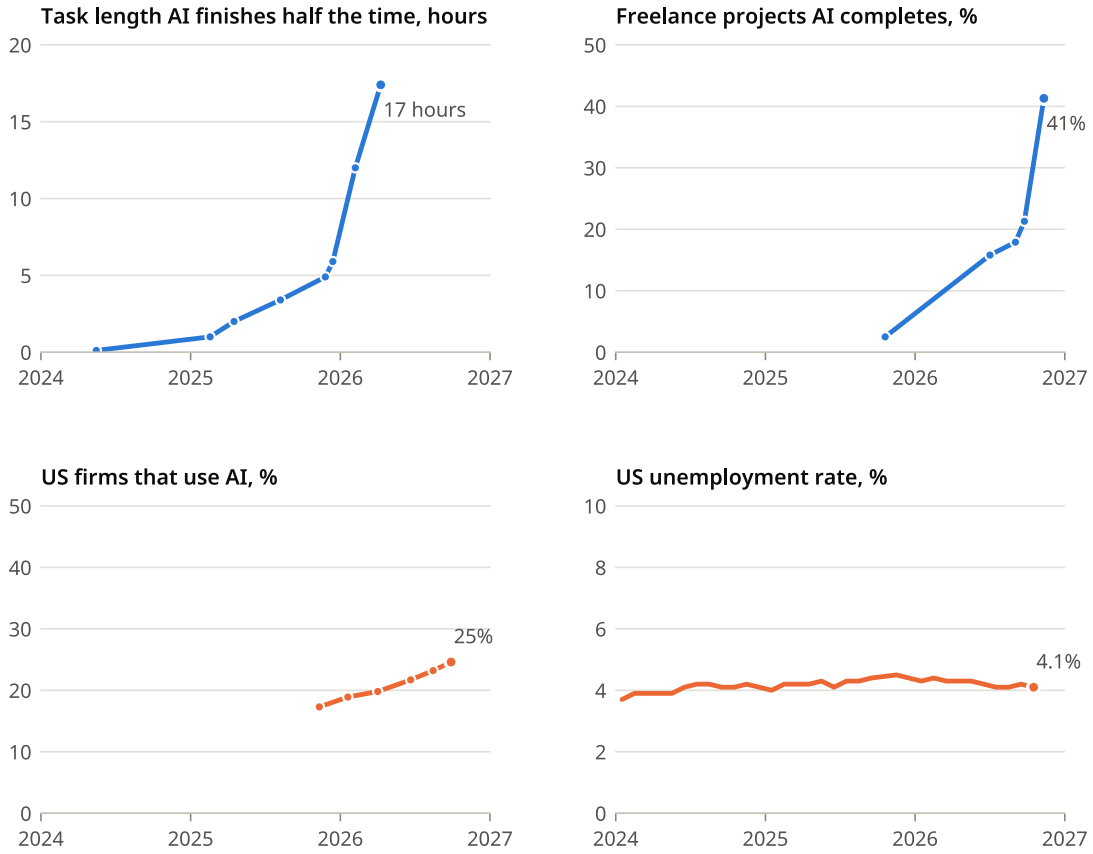


Figure 3. Two measures of what AI can do, and two measures of the economy around it. Top left: the length of software task, measured in the time a human expert needs, that the best AI system completes half the time. Top right: the share of real freelance projects that the best AI agent completes to the standard of the paid professional. Bottom left: the share of American firms that report using AI in any part of their business. Bottom right: the unemployment rate.
Source: METR; Center for AI Safety and Scale AI; US Census Bureau; US Bureau of Labor Statistics. `models/fig_gap.py`

What the machines can do

Mental work. The clearest independent yardstick has been the measure kept by the research group METR: the length of task that an AI system completes half the time. It was seven minutes for the best system of mid-2024, one hour in February 2025, five hours in November 2025 and seventeen hours in April 2026, a doubling roughly every four months [2]. The yardstick has since broken. METR's set of test tasks contains almost nothing longer than sixteen hours, the newest systems have learned to game it, and METR stopped updating its public chart in September 2026.

The best remaining public measure is the Remote Labor Index, which gives AI agents real freelance projects, in design, data analysis, video, software and so on, and has people grade the result against what the paid professional delivered. The best agent completed 2.5 percent of the projects in October 2025, 16 percent in July 2026 and 41 percent in November 2026 [3].

Against that, the limits are specific and well documented. On a new test of eight-hour professional tasks set in realistic working conditions, the best system passes 12 percent, and close to

none in finance and law, where a single mishandled figure spoils the work [4]. When the maintainers of real software projects reviewed AI-written changes that had passed the automated tests, they said they would reject about half. When researchers gave leading agents several days and a large budget to produce original research, the results were turned down by the scientists whose work they were extending. And the systems still do not learn on the job: an agent on its thousandth day at a task is no better than on its first [5].

One more development belongs here. AI is now doing a large part of the work of building AI. More than nine tenths of the computer code written at one leading laboratory is written by its own models, and METR's independent assessment is that this laboratory's progress is running at about twice the rate of the preceding two years, mostly for that reason [20]. The estimate is uncertain, rests on one company's partly withheld data, and comes out lower across the industry as a whole. If it holds, it is the first measured instance of the fifth property listed in Chapter 4: machine labor working on itself.

Physical work. Driving is done. Waymo carries more than 500,000 paying passengers a week in about fifteen American cities, with around 4,000 cars and no drivers, and over 271 million driverless miles its cars have been involved in about four fifths fewer injury crashes than human drivers. That is a real service, and it is still small beside the 4.3 million Americans who drive for a wage.

Everything else physical is much further back. Somewhere between 7,000 and 16,000 humanoid robots were sold worldwide in 2025 and perhaps 50,000 will be in 2026, nearly nine tenths of them Chinese, most of them to laboratories, schools and showrooms. The American company with the most real customer hours had revenue of $1.8 million in 2025. Tesla conceded in January 2026 that none of its robots was yet doing useful work in its own factories. The software is improving quickly from a low base: one company reported in September 2026 that its robot completed 56 percent of long household chores in thirty homes it had never seen, against 9 percent for the same robot without a new method of learning from video of people. An independent audit finds robots two to ten times slower than people at most handling tasks, and judges that almost all manipulation outside warehouses is still at the laboratory stage [6], [21], [22].

What machine work costs

Figure 4 sets the cost of an hour of work from a person beside the cost from a machine.

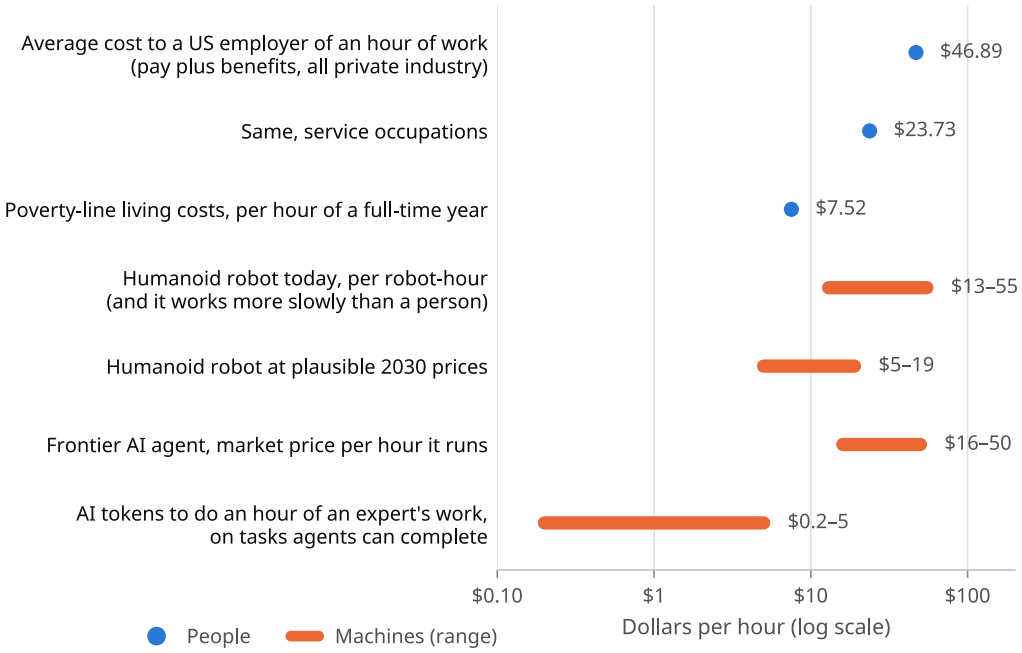


Figure 4. What an hour of work costs. Blue dots are people; orange bars are ranges for machines. The bottom row is the comparison that matters for the argument of Chapter 5: on tasks AI agents can complete, the computing cost of an expert's hour is already far below the poverty line. The rows above it show why that has not yet emptied offices or factories. A frontier agent is priced by the hour it runs at about the level of a service worker, and succeeds only some of the time. A humanoid robot still costs more per hour than the worker it would replace, and works more slowly.
Source: US Bureau of Labor Statistics (employer costs, June 2026); federal poverty guideline for one person; Epoch AI and METR (AI); Agility Robotics' filed prices and the estimates in research note 04 (robots). `models/fig_cost_of_work.py`

Three facts about cost are easy to confuse and all are true. The price of a *fixed* level of AI capability falls about tenfold a year: the small models of late 2026 cost a tenth of what the small models of late 2025 did and are better. The price of the *best* capability has not fallen, and has risen, because each laboratory has added a more expensive tier at the top. And the amount of computation used per task is rising fast, because agents that work for hours consume enormous quantities of it; the heaviest users inside the AI companies now run up bills for agents that are comparable to their own salaries [1], [5].

So "intelligence too cheap to meter" describes last year's intelligence. For the tasks that last year's systems could do, **P10** already holds by a wide margin. For the tasks at the frontier, the binding limits today are whether the machine can do the work at all, and how much human time goes into specifying and checking it.

The money

The revenue of the two leading AI companies taken together ran at about $30 billion a year in January 2026 and about $105 billion by late summer. One of them, Anthropic, reported in a filing with securities regulators that its revenue for the first nine months of 2026 was $33 billion, against $4.6 billion for the whole of 2025, and its run-rate at the end of October was about $81 billion [23]. Three quarters of that comes from fewer than twelve thousand customers. This is so far a business of intensive use by a narrow set of buyers, above all software teams.

The spending is much larger than the revenue. Amazon, Alphabet, Microsoft and Meta spent $376 billion on capital investment in 2025 and will spend more than $700 billion in 2026, most of it on AI data centers. By one estimate AI investment is running at about 3.6 percent of American output, a larger share than the railways took at their nineteenth-century peak. The cash is no longer coming from profits alone. Two of the four now spend more than they take in, the group will issue around a quarter of a trillion dollars of bonds this year, and more than a trillion dollars of further commitments sit off their balance sheets. A common rule of thumb puts the AI revenue needed to justify the build-out at about $1.6 trillion a year [19].

The strain is showing. Long-term interest rates are around 5 percent. On 5 November 2026, two days after the midterm elections, technology shares had their worst day in more than a year; the index of chip-makers is more than a third below its June peak. On the day this is written, the chief executive of the most important chip company has resigned. None of this shows that the technology has stalled, and demand for it still exceeds supply. It does show that the financing of the build-out can fail on its own schedule, and Part IV treats a financial bust as likely.

What has happened to work

The headline is that almost nothing has happened, and the detail is that something has.

The aggregate. The American unemployment rate was 4.1 percent in October 2026, and has been between 4.0 and 4.5 percent for two years. The share of people aged 25 to 54 who have a job is 80.7 percent, close to its highest level in a quarter of a century. Productivity is growing at about 2 percent a year, its long-run average. Careful estimates of how much AI has raised unemployment so far cluster around one tenth of a percentage point [18], [24], [25].

The pattern underneath. The labor market is weak in one particular way: employers have nearly stopped hiring. The rate at which firms take on new workers is well below its 2019 level, layoffs are low, and those who do lose a job are taking much longer to find another. The long-term unemployed are now 28 percent of all the unemployed, against 21 percent at the start of 2025.

Within that, the best payroll data show a specific gap. Among workers aged 22 to 25 in the occupations most exposed to AI, such as software development and customer support, employment is 19 percent below where it would be if it had followed the path of their peers in less exposed occupations, and the gap has widened in each release of the data. It has opened through hiring that did not happen, with no rise in dismissals [26]. The cause is disputed. Other data sources show a much smaller gap, and a study in Denmark found the same decline at firms that had not adopted AI at all [27]. But at the one AI company that has published figures on its own hiring, the share of technical recruits who are junior fell from 27 percent to 9 percent in under two years [28].

A handful of occupations have visibly shrunk. Between 2022 and 2025 the number of computer programmers fell by 31 percent, data-entry clerks by 19 percent, customer service representatives by 10 percent and writers and editors by about 10 percent; call centers have lost 27 percent of their jobs since late 2022. Several of these were shrinking before. Meanwhile the number of software developers, a broader and better-paid category, rose by 10 percent [18].

This is what **P9** predicts for the early part of a transition, and it is also what a skeptic predicts. Tasks are being taken over, the people who do what remains are doing well, and the cost falls on those who are not hired. It matches the order of events that the authors of "The Intelligence Curse" forecast in 2025: hiring freezes at the bottom first [29].

One long-running trend. Figure 5 shows the share of American business income paid for work. It is at its lowest since records began in 1947. The fall began around 2000, long before the present AI systems, and no study attributes the latest leg to them. It is a reminder that the wage system was losing ground as a way of distributing income before the premise of this guide came into view.

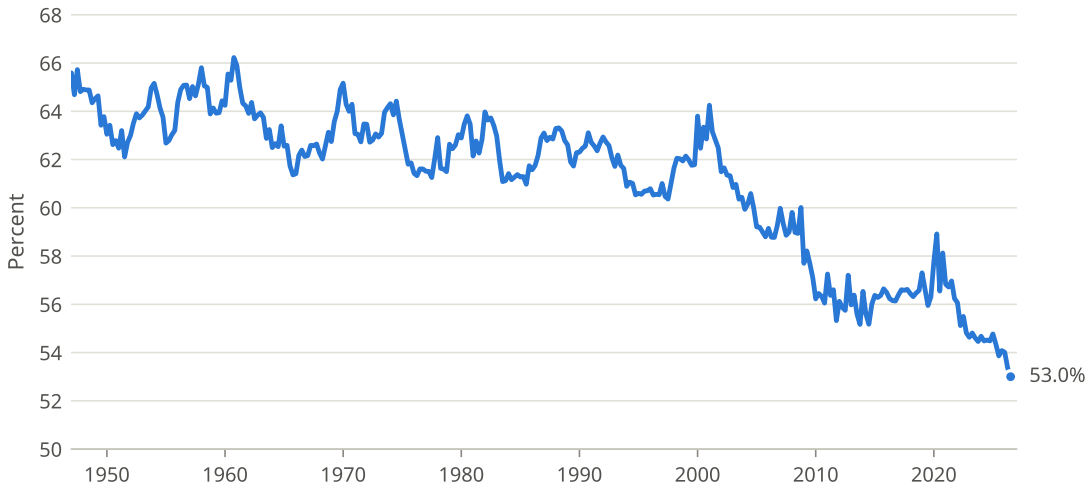


Figure 5. The share of US business income paid for work, 1947–2026. Nonfarm business sector, quarterly. The share was close to 64 percent for half a century, began to fall around 2000, and reached 53.0 percent in the third quarter of 2026, the lowest on record.
Source: US Bureau of Labor Statistics via FRED, series PRS85006173, scaled to the published level for Q3 2026. `models/fig_labor_share.py`

What governments have done

The United States has enacted no federal policy aimed at job loss from AI. The one large law of the period, the budget act of July 2025, went the other way, cutting more than a trillion dollars over ten years from medical and food assistance for low-income households. Federal AI policy has concentrated on speeding the build-out and on stopping the states from regulating it.

The politics, though, moved a long way in 2026 [10].

- A wave of bills arrived in Congress, none yet voted on: a public fund holding half the shares of large AI companies and paying everyone a dividend; taxes on AI usage and on excess profits to pay for jobs and for displaced workers; unemployment benefits that lengthen automatically when unemployment rises; a requirement that large firms report layoffs caused by AI.
- The idea of public ownership came from both sides within days of each other. In June 2026 President Trump said the government might take stakes in the leading AI companies, and Senator Sanders introduced his bill. One of the companies was reported to have offered the

government 5 percent of itself. All three of the largest American AI companies have now said in writing that the public should have some share of the returns.

- In September the President promised every adult $5,000 if his party kept control of Congress. It did not. On 3 November 2026 the Democrats won both chambers, and the government is divided until at least January 2029.
- The anger that showed up at the ballot box was about the physical footprint of AI. Voters took the side of restricting data centers in forty of forty-one local measures, and the governors-elect of four states that host about half of the planned large sites campaigned on pauses or conditions [30].
- The states passed 115 laws on AI in 2026, almost all about disclosure and procedure. None taxes robots or AI. Trade unions with a physical chokepoint, the dockworkers above all, won bans on automation; the rest won the right to be told.

Public opinion is lopsided and not simple. Between 71 and 84 percent of Americans, depending on the poll, expect AI to cost jobs [31]. But the last clean poll on a universal basic income found a majority against, and in 2026 people preferred support that is earned to unconditional cash by a wide margin, while two thirds favored a "dividend" paid from the profits of AI companies. The words matter.

Reading the evidence

Who has been right so far? The research note on the landscape of forecasts scores the dated predictions of the main camps, and the result is instructive [32].

On the technology, the money and the investment, the people who expected very rapid progress have been closest, and nearly everyone else was too low. Forecasts of a plateau in capability, of weak demand, and of a bust in 2025 all failed.

On unemployment, growth and productivity, the gradualists have been right. Those who predicted mass white-collar job loss within months, or an economy visibly transformed by 2026, were wrong.

The forecasts that did best held both views at once: *capability arrives quickly, and its effects on the economy arrive slowly*. One written in 2021 described 2026 as the year when AI assistants would finally work for serious economic tasks while world output showed no acceleration at all. That is close to exact.

I take that as the working rule for the chapters that follow, with one warning attached. The gradualists have so far been tested only in the period when both camps predicted little visible change. The distinctive claims of the fast camp fall due between 2027 and 2030.

WHERE THE CHAIN STANDS

- Capability is advancing at a pace that has broken its own yardsticks. Reliability on long, messy, high-stakes work is the main thing still missing. Robots are a decade behind software, except in driving.
- On tasks machines can do, machine cost is already far below a person's upkeep. At the frontier it is not, yet.
- The build-out is the largest investment boom in a century and is financed in a way that can break.
- Employment, wages and productivity look normal in aggregate. The visible effects are in hiring, at the entry level, in a few occupations.
- Policy has not acted, but the terms of the debate, including public ownership, shifted sharply in 2026.
- The evidence supports one rule: capability outruns diffusion.

CHAPTER 12

Minds first, hands later

The premise has two halves, mental labor and physical labor, and the evidence of the last chapter says they are arriving a decade apart. This matters more than any other fact about the next ten years, because an economy in which machines can think but cannot yet handle the physical world behaves differently from one in which they can do both. This chapter takes the two halves separately, then puts them together.

Why the two halves move at different speeds

The difference is not that physical tasks are harder to learn, though many are. It is that mental machine labor is made of something that already exists in bulk, and physical machine labor is not.

A machine mind is a program running on chips in a data center. The chips are being manufactured in tens of millions, and a program that can do a job can be started a million times over on the day it is ready. A machine body has to be built, one for each worker replaced, out of motors, gears, sensors and batteries, in factories that mostly do not exist yet. Figure 6 shows the result.

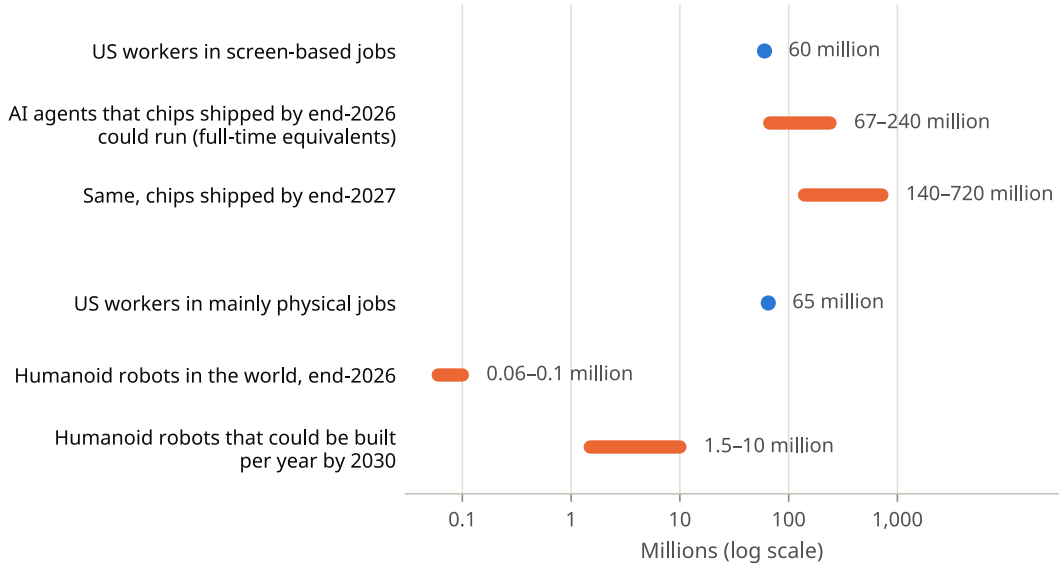


Figure 6. Machine minds are plentiful; machine hands are not. The upper group compares the number of Americans in screen-based jobs with the number of full-time AI agents that already-manufactured chips could run, if all of them were used for that purpose and for tasks that agents can do. The lower group compares Americans in mainly physical jobs with the world's stock of humanoid robots and with the number that could be built each year by 2030 under strong demand.

Source: Epoch AI (October and April 2026); US Bureau of Labor Statistics; Dingel and Neiman (2020); industry trackers summarized in research note 04. `models/fig_capacity.py`

The chips shipped by the end of 2026 could run the equivalent of somewhere between 67 million and 240 million full-time agents; those shipped by the end of 2027, between 140 million and 720 million [33]. About 60 million Americans work in jobs that can be done entirely at a computer [34]. So for mental work, the quantity of machine labor is not the constraint and will not become one. What limits substitution is whether the machines can do the work reliably, and how fast organizations change. When they can, the supply is there, and **P8** says the price of that work will fall to its cost quickly.

For physical work the constraint is quantity as well as capability. The world's stock of humanoid robots is under a hundred thousand. The most careful study of the supply chain concludes that 1.5 to 3 million a year could be built by the end of 2030 if buyers wanted them, and perhaps 5 to 10 million with an aggressive expansion of the factories that make precision gears [35]. There are 65 million Americans in mainly physical jobs, and several times that number in the rest of the rich world. Even on the fast path, robot labor stays scarce for years after it becomes capable, and scarce machine labor is priced above its cost. Physical wages therefore hold up longer.

The first transition: work done at a screen

Take the three kinds of work that the scenario model in Chapter 14 uses. Screen-based work is about 37 percent of American jobs and, because it is better paid, about 46 percent of all wages. It is where the highest-paid tenth of workers are concentrated, and it is the main occupation of the large metropolitan areas and of people with university degrees.

Three things follow from the earlier chapters about how this work changes.

It goes task by task, and the last tasks matter most. By **P9**, a profession in which machines do 80 percent of the tasks can be a better-paid profession than before, for the people who do the other 20 percent, which are typically the tasks of deciding what is wanted, checking the result and answering for it. The evidence of 2026 fits this: fewer junior recruits, higher pay for experienced staff, and rising output. The profession reaches the cliff when the machine can also specify, check and answer for the work.

Entry closes before exit opens. The adjustment so far is through hiring. Firms do not dismiss the people they have. They stop taking on beginners, whose work is the first that agents can do. For a time this protects everyone already inside. It also means that a profession can be dying while nobody in it has lost a job, and that the cost falls first on the young. The supply of experienced people, who learned by doing the junior work, then begins to dry up, which increases the pressure to automate the senior work too.

It reverses forty years of pattern. Since about 1980, new technology has favored the educated over the uneducated, the office over the factory, and the big city over the small town. The first transition runs the other way. The plumber and the nursing aide are, for a decade, safer than the paralegal and the financial analyst. This has no precedent in living memory, and Chapter 9 argued that its political effects will be large because of who is affected.

The refuge: work done with hands

About 42 percent of American jobs, with 29 percent of wages, are mainly physical: transport, food service, production, construction, repair, cleaning, care assistance. During the first transition this work is the bottleneck, and **P4** applies to it in full.

There will be more of it to do. A world with abundant machine minds designs more buildings, plans more infrastructure and invents more products, all of which still have to be made, installed and maintained by hand. The data-center and power build-out alone is already adding construction work. And machine minds can direct human hands: a worker wearing a camera and an earpiece, guided step by step by a system that has absorbed every repair manual ever written, can do work that used to need years of training.

This is good news for physical workers, with two qualifications.

The first is crowding. Displaced office workers do not vanish. Many will move into physical and in-person work, and their arrival holds wages down. The last time a large class of routine office and factory jobs disappeared, between 1980 and 2010, the people displaced moved mostly into lower-paid service work, and wages there stagnated for a generation.

The second is that machine guidance cuts both ways. If a system can talk a novice through a master electrician's work, the master's skill is worth less. Physical work as a whole may be in demand while the premium for experience within it shrinks.

So the realistic picture of the refuge is plenty of physical work, at wages that rise modestly for the least skilled and fall for the most skilled, with far more people competing for it. It is a refuge, and it is temporary.

The second transition: robots

The refuge ends when general-purpose robots are both capable and numerous. Each of these has its own clock.

Capable. The gap between a robot that completes a household chore 56 percent of the time and one that does so 99.9 percent of the time is the same kind of gap that driverless cars took about ten years to cross, from the first rides without a safety driver to a commercial service. Robots have advantages the cars did not: far more investment, far better learning methods, and, soon, machine minds doing the engineering. My judgment is that reliable general-purpose robots for structured settings, such as warehouses, factories and commercial kitchens, arrive around 2029 to 2032, and for unstructured ones, such as a crawl space under an old house or the bedside of a frail patient, in the middle of the 2030s or later.

Numerous. From tens of thousands to millions a year by 2030 is feasible. Tens of millions a year by the mid-2030s would require the kind of manufacturing effort that built the car industry, compressed into a decade, with most of the supply chain presently in China. It can be done; whether it is done by 2036 is a question of money and politics more than of physics.

Cheap. Figure 4 showed a humanoid robot costing $13 to $55 an hour today. At plausible 2030 prices the figure is $5 to $19, against $24 to $48 an hour for the American workers it would replace. The larger part of the robot's cost at that point is its share of a human supervisor. The moment that matters is when one person can oversee dozens of robots, or none is needed.

The order in which physical work goes is fairly predictable from how structured the setting is and how costly a mistake would be. Driving and warehouse handling go first, and are already going. Factory work, cleaning, stocking shelves and fixed-menu cooking follow. Delivery to the door, farm work and hotel housekeeping come later. The skilled trades in existing buildings,

hands-on care of the old and the sick, and emergency work come last, and some of that last group, being work where the person is the point, does not go at all (**D5**).

If only one half arrived

It clarifies the argument to ask what would happen if only one of the two transitions took place.

Minds only. Suppose machines could do all mental work and robotics stalled for a generation. The economy would be limited by its hands. Growth would be faster than today but far from explosive, because every plan still has to be carried out by people (Chapter 6). The share of income going to physical and in-person work would rise. Wages would not collapse, because a necessary task would remain that only people could do. But the wage structure would be compressed toward the pay of manual work, the premium for education would largely disappear, and labor's total share of income would fall, since nearly half the wage bill is now earned at a screen. That world is recognizably ours, harsher for the educated and somewhat kinder to manual workers, with a much richer class of owners. It is roughly what the central scenario of Chapter 14 looks like in 2036.

Hands only. Suppose instead that robots could do all physical work but machines could not think. This is the twentieth century continued: machines take the muscle, people supply the judgment, and wages rise. It is also impossible, since a robot that can do all physical work must be able to perceive, plan and adapt, which is to say think.

Both. With both, there is no refuge, and Parts II and III apply without qualification.

The point of the exercise is that *the second transition is the one that changes the kind of economy we live in.* The first redistributes income and status within a wage economy. The second ends the wage economy. And the first speeds the second, because the scarcest input to robotics has been engineering talent, and engineering is mental work.

WHERE THE CHAIN STANDS

- Mental machine labor can be supplied in unlimited quantity as soon as it is capable. Physical machine labor has to be manufactured, body by body.
- Screen-based work, 37 percent of jobs and 46 percent of wages, is therefore exposed first and fastest. The adjustment comes through hiring before it comes through layoffs.
- Physical work is a refuge for roughly a decade, with more work, more competition for it and less reward for skill.
- Robots end the refuge when they are capable, numerous and cheap: beginning in the early 2030s in structured settings and finishing later, if ever, in the least structured.
- The first transition reshapes the wage economy. The second ends it.

CHAPTER 13

What sets the pace

The argument of Parts II and III tells us where the premise leads. It does not tell us when. The speed of the transition is governed by five separate clocks, which run at very different rates, and most of the confusion in public debate comes from reading one clock and assuming it sets the time for all.

Five clocks

The capability clock runs in months. The length of task that AI systems can complete has been doubling about every four months. The share of real freelance projects they complete rose from 16 percent to 41 percent in four months of 2026. This is the clock the AI laboratories watch, and by it the premise for mental work is a small number of years away.

The diffusion clock runs in years. The share of American firms using AI at all rose from 17 percent to 24 percent in the year to October 2026, which is steady and unspectacular. About 40 percent of workers use it for something, a figure that has not moved in a year, and it touches about 5 percent of all hours worked [36], [37]. Executives at nine thousand firms report that it has raised their productivity by 0.2 percent a year [38].

History says this clock is slow for good reasons. The technologies that reorganized production, as opposed to those that households simply bought, took 25 to 45 years to go from a tenth of their market to nearly all of it: the power loom 25, the electric motor in factories about 30, the tractor 39 [39]. A firm has to redesign its processes, retrain or replace its people, settle who is liable for errors, and persuade customers and regulators. The old method also fights back by getting cheaper. The falling price of horses kept them in use on American farms for years after the tractor was the better buy.

There are also good reasons to expect this technology to diffuse faster than those. It arrives over a network that already reaches every desk. The cost gap is enormous. It can do much of the work of installing itself. And a new firm built around machine labor from the start does not have to change anything; it only has to take customers from firms that are slower. My estimate is that the main body of screen-based work that is not protected by law takes eight to fifteen years to move from “the machine can do it” to “the machine mostly does it.” That is two to four times faster than the historical norm, and still far slower than the capability clock.

The build-out clock runs in years to decades. A data center takes one to three years to build. A connection to the electricity grid takes about five. Gas turbines ordered today are delivered in the early 2030s. A chip factory takes three to five years, and a mine sixteen or more [19]. For mental work this clock no longer binds, as Chapter 12 showed. For physical work it is the main one.

The money clock can stop. The build-out is running on borrowed money and on the expectation of revenue that is still mostly in the future (Chapter 11). Booms of this kind, in railways

in the 1840s and in fiber-optic cable in the 1990s, ended in a financial bust that ruined many investors and paused construction for some years, while the technology itself carried on and the assets built in the boom were used by whoever bought them cheaply. I expect the same here. A bust would slow the frontier for a year or three. It would not reverse what the existing machines can do, and it would arrive as a recession.

The political clock runs on elections and emergencies. American federal elections come every two years. Major social legislation has historically come three to six years after the need became undeniable: the banking and relief acts of the New Deal three and a half years after the crash of 1929, Social Security nearly six. Emergency money can move much faster: in March 2020 Congress passed $2.2 trillion of relief fourteen days after the emergency was declared, and the poverty rate *fell* during a period of mass layoffs. But what is passed in an emergency tends to expire with it, as that relief did [39].

How displacement actually happens

Put the clocks together and a particular sequence follows. It is worth spelling out, because it is unlike the popular image of robots marching into offices.

1. **Capability arrives and nothing visible happens.** Firms experiment. Their existing staff use the tools and become more productive. Output rises a little.
2. **Hiring stops.** Managers find they do not need the next recruit. Nobody is dismissed, so nothing appears in the layoff statistics. This is where the United States is in 2026 (Chapter 11).
3. **Attrition does the cutting.** About one American worker in thirty leaves their job every month for ordinary reasons. A firm that simply does not replace them shrinks by a tenth or more a year without a single dismissal.
4. **A recession does the rest.** Firms cut deeply in downturns and reorganize when they rebuild. In each of the last three American recessions, the routine jobs that were lost did not come back in the recovery [40]. The next recession, whatever causes it, will be the moment when firms rebuild around machine labor. Output will recover, and employment in the affected work will not.
5. **The statistics catch up.** Only at this stage does the change appear in the numbers that politics responds to.

The fourth step is the important one. It means the transition will be recorded in history as a slump followed by a jobless recovery, and many people will say the slump caused the job losses. It also means that the long quiet period in which capability runs ahead of the statistics is misleading. Displacement is being stored up, and a downturn releases it.

Forecasts

Everything up to this point in the guide has been either a fact or a deduction. From here on I have to make judgments about the future, and I will mark them as such and attach probabilities. A probability of 60 percent means that I expect to be wrong about four such statements in ten. These are my own estimates, formed from the evidence in the research notes. The “80 percent range” given with some of them is the interval I think has four chances in five of containing the outcome.

FORECAST 1 When machines can do the screen-based work
AI systems will be able to do at least nine tenths of the tasks in screen-based jobs as well as a competent person, at lower cost, by the end of 2029 with probability 55 percent, and by the end of 2032 with probability 80 percent.

The case for sooner is the measured trend and the measured acceleration of AI research by AI. The case for later is that reliability, judgment and learning on the job have improved more slowly than raw task length, and that a financial bust may slow the frontier. The authors of the best-known fast scenario now put their median for AI that outperforms top experts at everything in late 2027 or 2028; a large panel of forecasters puts the arrival of general AI around 2050 [41], [42]. I am between them and closer to the first.

FORECAST 2 How much screen-based work machines actually do by 2036
In 2036, machines will be doing about 45 percent of the screen-based work now done by Americans (80 percent range: 15 to 80 percent).

FORECAST 3 How much physical work robots do by 2036
In 2036, general-purpose robots and driverless vehicles will be doing about 8 percent of the physical work now done by Americans (80 percent range: 2 to 30 percent). Driving will be the largest part.

FORECAST 4 The recession
A recession will begin in the United States before the end of 2029 with probability 65 percent. If it does, employment in screen-based occupations will fail to return to its previous peak in the recovery, with probability 70 percent.

The first number is only a little above the historical frequency of recessions, raised for the present combination of high interest rates, an inflation problem and a debt-financed investment boom. The second is the claim that matters.

FORECAST 5 Employment, labor's share and growth in 2036
- The share of Americans aged 25 to 54 who have a job, now 80.7 percent, will be about 75 percent (80 percent range: 62 to 80).
- The share of business income paid for work, now 53.0 percent, will be about 47 percent (80 percent range: 36 to 52).
- Output will have grown by an average of about 3 percent a year over the decade, after inflation (80 percent range: 1.8 to 6). The chance of any single year of growth above 10 percent by 2036 is about one in eight.

FORECAST 6 **The policy response by the early 2030s**

- The federal government will hold shares in at least one leading AI company by 2030: 60 percent.
- A federal law will create a regular payment to all adults, or a dividend, financed from taxes on AI or capital or from public shareholdings, by 2032: 45 percent.
- Such a payment will be worth at least $3,000 a year per adult by 2036: 15 percent.

These numbers are the skeleton of the scenarios in the next chapter. A reader who disagrees with any of them can trace exactly where their own forecast would part from mine.

WHERE THE CHAIN STANDS

- Capability moves in months, diffusion in years, physical build-out in years to decades. Money can halt the build-out for a time. Politics moves at the speed of elections, or of emergencies.
- Displacement comes as a hiring freeze, then attrition, then a recession from which the affected jobs do not return.
- My central estimates for 2036: machines doing a little under half of today's screen-based work and under a tenth of physical work; the prime-age employment rate down about six points; labor's share down about six points; growth a point above trend.
- The policy response will probably include public stakes and some form of dividend, and will probably be small.

CHAPTER 14

Six scenarios for 2026 to 2036

A scenario is a consistent story about how the open questions resolve. The open questions here are few. Part IV has reduced them to three: how soon machines can do the work, mental and then physical (F1); how fast that capability is put to use (F2, F3); and what the political system does about it (F6).

The six scenarios below cover the combinations I think deserve weight. Five of them are distinguished mainly by the pace of the technology, and for those I have run the numbers through a small accounting model so that each story adds up. The sixth, "Shared dividend," follows the same technological path as the central scenario and differs in the political response.

The model is described in Appendix A. Its job is consistency: it takes an assumption about how much of each kind of work machines do in each year and returns the hours of human work, the employment rate, labor's share of income and the level of output that go with it. Its relations are calibrated to reproduce the three scenarios for 2030 published in September 2026 by a group of economists that included Anton Korinek and Charles Jones [43]. It is bookkeeping for stories, and its outputs should be read as round numbers. Figure 7 shows the five paths, and Table 3 compares all six in 2036.

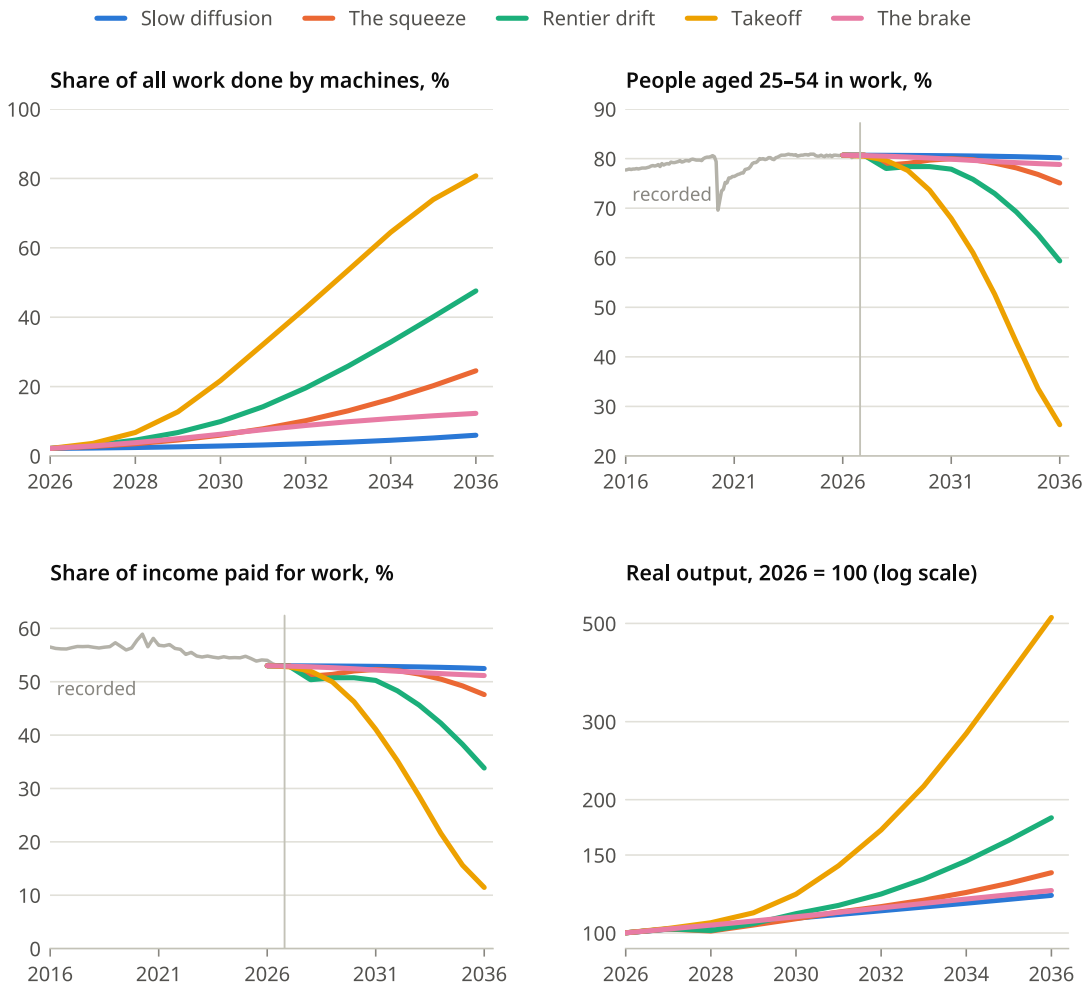


Figure 7. Five paths through the decade. Each line is one scenario's assumption about how quickly machines take over work (top left) and what the accounting model says follows for employment, labor's share of income and output. Gray lines show the recorded past. The dip in 2028 in two of the scenarios is an assumed recession. "Shared dividend" follows the same lines as "The squeeze" and differs in how income is distributed.

Source: `models/scenarios.py`, `models/fig_scenarios.py`. History from the US Bureau of Labor Statistics via FRED.

Table 3. The six scenarios in 2036. Machine shares are the scenario's assumptions; the last three columns are outputs of the accounting model. *In "Shared dividend" a shorter standard working week spreads the same hours of work over more people, so the employment rate would be higher than shown.

	Scenario	My odds	Screen work done by machines	Physical work done by machines	Aged 25–54 in work	Labor share	Growth a year
	Today (2026)		5%	under 1%	80.7%	53.0%	about 2%
1.	Slow diffusion	15%	13%	2%	80%	52%	2.0%
2.	The squeeze	40%	51%	10%	75%	48%	3.2%
3.	Shared dividend	12%	51%	10%	75%*	48%	3.2%
4.	Rentier drift	13%	81%	34%	59%	34%	6.2%
5.	Take-off	12%	98%	84%	26%	12%	18%
6.	The brake	8%	28%	4%	79%	51%	2.2%

1. Slow diffusion

AI turns out to be a normal technology, and normal technologies take decades.

What has to be true. Either capability stops short of the premise, with reliability, judgment and learning on the job proving much harder than the trend suggests, or capability arrives and the ordinary frictions of organizations, liability and regulation hold adoption to the pace of electricity and the tractor.

How it unfolds. The investment boom ends in a bust around 2027 or 2028, and this time the bust does damage to the frontier: the laboratories cannot fund the next generation of systems. AI settles in as a very good tool. Programmers, analysts and writers use it all day and produce more. Firms adopt it over twenty years. Employment in a few occupations shrinks, as it did for typists and travel agents, and others grow. In 2036 the economy is a few percent larger than it would have been and the arguments of this guide read as premature.

For and against. This is what almost every official forecaster expects, and by their own measures they have been right so far [32]. The economist Daron Acemoglu puts the total gain at about 1 percent of output over a decade [44], and the scholars who coined the phrase "normal technology" make the case from the history of diffusion [45]. Against it stands the whole capability record of Chapter 11. To believe this scenario, one has to believe that a trend which has held for seven years and is now accelerating will stop within about two years, short of a

destination that is a few doublings away. One also has to explain why firms would be slow to adopt something that does the work for a few percent of the cost.

My probability: 15 percent.

2. The squeeze

The machines arrive for mental work at the speed of diffusion, and the United States responds late and in pieces.

What has to be true. F1 comes true around 2029 or 2030. Adoption takes roughly a decade. Robots improve steadily but are still scarce in 2036. Politics produces a patchwork.

How it unfolds. The late 2020s look like 2026, only more so: hiring of beginners in screen-based work dries up, experienced staff are well paid, and output per worker climbs. Somewhere between 2027 and 2029 a recession arrives, set off by the financing of the AI build-out, by interest rates or by something unrelated. Firms cut their office staff deeply and rebuild around agents. The recovery, when it comes, is the most jobless on record. Unemployment peaks in the high single digits, and then, more tellingly, people stop looking: the share of adults in work drifts down year after year.

The pain falls on the professional class and on the young who expected to join it. Law, finance, software, media, marketing, consulting and administration all shrink. Physical and in-person work absorbs some of the displaced, at lower pay. The trades do well. So do the owners of shares, land and energy.

Washington responds the way it usually does. Unemployment benefits are extended, then extended again. There are rounds of checks, called dividends. The government takes shares in the AI companies, partly as a condition of the support it gives them in the bust. There are new taxes on AI usage and on data centers, set too low to matter. States pass laws requiring a human in a widening list of roles. Social Security's trust fund, due to run short in the early 2030s, forces a larger bargain, and some money from the machine economy is routed into it.

In 2036. Machines do about half of what was screen-based work and a tenth of physical work. Three quarters of people aged 25 to 54 are in work, down from four fifths. Labor's share of income is 47 or 48 percent. Total pay has roughly kept pace with a no-AI economy, but it is shared among fewer people and the premium for education has shrunk. The economy has grown a little over 3 percent a year. Robots are being built by the million, and everyone can see what comes next. The country is perhaps a third of the way through the transition, and has not decided who will own the result.

For and against. This scenario is the direct application of the rule that capability outruns diffusion, which is the rule the evidence supports. Its technological assumptions are my median forecasts. Its political assumptions follow from the present line-up: a public that expects harm from AI and prefers earned support, both parties interested in public stakes and dividends, and a system that acts in emergencies and lets emergency measures lapse. The main argument against it is that it sits between two more self-consistent stories. If AI really accelerates its own development, as the measurements of 2026 suggest, there is little reason for the pace to stay moderate (scenarios 4 and 5). And if adoption is slow enough to stretch over a decade, perhaps it is slow enough to stretch over three (scenario 1).

My probability: 40 percent.

3. Shared dividend

The same technology as the squeeze, met by a durable public claim on the income of the machine economy.

What has to be true. The recession and the jobless recovery produce a political majority for a structural response, and the response is built on ownership and automatic entitlement, the durable forms identified in **P27**, while the majority still has the leverage to demand it (**P24**).

How it unfolds. The election of 2028 is fought on AI and jobs. The winning coalition takes the public-stake idea that both parties floated in 2026 and makes it large: a national fund that receives a share of the equity of the largest AI and computing companies, a levy on data-center capacity and energy sites, and a mandate to pay every adult an equal dividend. Unemployment insurance is rebuilt to switch on automatically. Health insurance is separated from employment. The standard working week is shortened. A national service program offers paid work in care, conservation and teaching to anyone who wants it.

In 2036. Employment, labor's share and output are as in the squeeze, but the dividend is a few thousand dollars per adult and rising with the machine economy, the fund owns a tenth or more of the AI sector, and the principle that every citizen has a property claim on machine output is established in law before robots arrive in force. That last fact is what distinguishes this scenario from all the others.

For and against. For: the idea already has sponsors in both parties and in the industry itself, the phrase "dividend" polls well, the first victims are politically powerful, and the Social Security shortfall creates an occasion. Against: the United States has passed structural social legislation on this scale twice in a century, both times with one party holding very large majorities, and has let its recent emergency programs expire. The sums are an order of magnitude beyond anything proposed with a number attached (Chapter 10). The industry's support is for a small stake. And the public's present anger is aimed at data centers, which leads to restricting the machines, a different thing from owning them.

My probability: 12 percent. I think a small version of this, a token dividend and a stake of a few percent, is more likely than not; that version belongs to the squeeze. The 12 percent is for the durable version.

4. Rentier drift

The technology moves fast, robots follow closely, and the gains are captured by those who own the machines.

What has to be true. **F1** comes true by about 2028. AI research accelerates further and diffusion is fast, because machine labor installs itself and new firms built around it take markets from old ones. Capable robots arrive around 2030 and are built by the tens of millions by 2036. The political response stays small.

How it unfolds. The sequence is the same as the squeeze, compressed and carried further. By the early 2030s most screen-based work is done by machines, and by the middle of the decade

physical work is going too. Output grows at 5 to 8 percent a year. Corporate profits and the value of land, energy and the leading firms rise to levels without precedent.

Politics does respond, but in the least durable ways. There is a means-tested income for the displaced, set near the poverty line and argued over every year. Jobs are protected by statute in health, education, law and government, so that a large part of the remaining workforce holds positions that exist by rule. The AI companies, now the largest taxpayers and the largest donors, are consulted on everything. Because machine-made goods are cheap, nobody starves, and the cheapness is offered as the answer to every complaint. Housing in the places people want to live is dearer than ever.

In 2036. Machines do four fifths of screen-based work and a third of physical work. Fewer than three in five people aged 25 to 54 have a job, and many of those jobs are protected ones. Labor's share of income is about a third. The economy is 80 percent larger than in 2026. The wealthiest tenth of households, who owned two thirds of the wealth at the start, own a good deal more.

For and against. This is the outcome that the logic of Part III produces if nothing intervenes: income follows ownership (P16), and leverage declines with need (P23). It is what the authors of "The Intelligence Curse" and "Gradual Disempowerment" warn of [29], [46]. The fast technological path has the capability evidence on its side. Against it: fast diffusion is the half of the fast story that the evidence has *not* supported so far, and robots at that scale by 2036 require nearly everything in manufacturing and politics to go right. And a faster, harsher transition seems more likely to provoke a large political response than a slow one, which pushes probability toward scenarios 3 and 6.

My probability: 13 percent for reaching this state by 2036. As a description of where the squeeze leads in the 2040s if nothing durable is built, I think it is the default.

5. Takeoff

AI improves itself, capability passes human level in everything within a few years, and the economy begins to grow at rates never seen.

What has to be true. The feedback loop measured at about two-fold in 2026 strengthens instead of fading. By 2028 or so, AI systems do AI research better than people. Systems far more capable than any person follow within a year or two. They design the robots and the factories, and physical production begins to double every year or two.

How it unfolds. I will not pretend to narrate this in detail. In the best-known account of such a path, the economy is transformed after the leap in machine capability, in special zones where regulation is lifted, with a million robots a month by the end of the decade [47]. In the accounting model, output in 2036 is five times its 2026 level, a quarter of prime-age adults are in what can still be called work, and labor's share of income is near a tenth.

What matters in this scenario is not economics. An economy growing at 20 percent a year can afford anything. The questions are whether the systems do what their owners intend, who the owners are, and whether any government is still in a position to tax and redistribute. P24 says that the majority's leverage is greatest before automation is complete. In a takeoff, the window is a few years long, and it is open now.

For and against. For: the measured acceleration of AI research by AI, the forecasting record of the people who expect this, and formal models in which a modest degree of automation across the research sectors is enough to tip growth into explosion [48]. Against: physical bottlenecks, the weak-link structure of the economy in which the slowest necessary step sets the pace [49], the absence of any sign of this in interest rates, which remain near 3 percent after inflation when a market that expected explosive growth would demand far more [50], and the fact that no large economy has ever sustained growth above about 19 percent, and that only for three wartime years.

My probability: 12 percent that growth of this kind is under way by 2036.

6. The brake

The build-out is stopped: by politics, by an accident or by a war.

What has to be true. One of three things. A political movement against AI wins: the local votes against data centers in November 2026 grow into state and then federal moratoria, and the leading countries, including China, reach some understanding that neither will race. Or an AI system causes a disaster, in finance or infrastructure or through a weapon, serious enough that strict controls follow, as they did for nuclear power after 1979 and 1986. Or the supply of advanced chips is cut off, most plausibly by a conflict over Taiwan, where nearly all of them are made.

How it unfolds. Capability already in existence stays in use, so screen-based work continues to change slowly. The frontier stops or crawls. Robots do not arrive in numbers. The economy of 2036 looks much like that of slow diffusion, reached by a different road, with the difference that the capability is known to be available and is being held back by decision.

For and against. For: the public is strongly negative about AI, the incidents of 2026 show that serious accidents are possible, and the most capable systems are already being withheld from general release on safety grounds. Against: the commercial and military incentives to continue are enormous, a brake in one country is an advantage to its rivals, and the technology runs on hardware that already exists in tens of millions of units. A brake by treaty would be one of the hardest feats of coordination ever attempted. A brake by war or disaster needs no coordination, which is why most of this probability sits there.

My probability: 8 percent.

Events that cut across the scenarios

Three events are likely enough, and independent enough of the scenario, to plan around.

A financial bust in the AI build-out (about 60 percent by 2029). It delays the frontier by a year or three, hands the data centers to new owners at a discount, and serves as the recession that triggers the jobless recovery. It makes slow diffusion more likely only if it starves the leading laboratories of funds for long.

The Social Security shortfall (near certain, early 2030s). The program's trust fund is financed by a tax on wages. Displacement brings its exhaustion forward. Because the program is the most popular in American government, Congress will be forced to find other money for it at

exactly the moment the machine economy is the obvious place to look. This is the likeliest occasion for the first large transfer from capital to households, and it will set a pattern.

A shock to chip supply (perhaps 10 to 15 percent over the decade). It would move any scenario toward the brake for several years.

What to watch

Each scenario implies observations that would be surprising under the others. These are the ones I would track.

Table 4. Signposts. None is decisive alone.

If you see this	it favors
No public AI system completes week-long, loosely specified projects by the end of 2028	Slow diffusion
A laboratory's research staff shrinks while its progress speeds up; interest rates after inflation above 4 percent	Takeoff, Rentier drift
Total employment in computer, legal, business and office occupations falls two years running outside a recession	The squeeze or faster
Unemployment among graduates aged 22 to 27 above 8 percent	The squeeze or faster
More than a million humanoid robots sold in a year before 2030	Rentier drift, Takeoff
Labor's share of business income below 50 percent before 2030	Rentier drift, Takeoff
Congress creates a fund that holds AI equity for citizens, with a dividend written into law, by 2030	Shared dividend
Emergency checks and benefit extensions that expire on schedule	The squeeze, Rentier drift
Moratoria on data centers in several large states; an AI accident with deaths or large financial loss	The brake

WHERE THE CHAIN STANDS

- The six scenarios differ in three things: when machines can do the work, how fast they are put to it, and what politics does.
- The central case, at 40 percent, is a decade in which mental work is squeezed, physical work is a temporary refuge, and policy is a patchwork.
- Two scenarios are much faster (25 percent together), two are slower (23 percent together), and one has the central technology with a durable political settlement (12 percent).
- In every scenario except the two slow ones, the question of who owns the machine economy is open in 2036 and more urgent than it is today.

CHAPTER 15

What I actually believe

The earlier chapters kept three things apart: what is true of any economy, what follows by logic from the premise, and what is a forecast. This chapter is all forecast. The guide was commissioned to end with my best guess and my real belief. Here it is, stated as plainly as I can, with the confidence I actually have. Figure 8 shows how I divide my belief among the six scenarios.

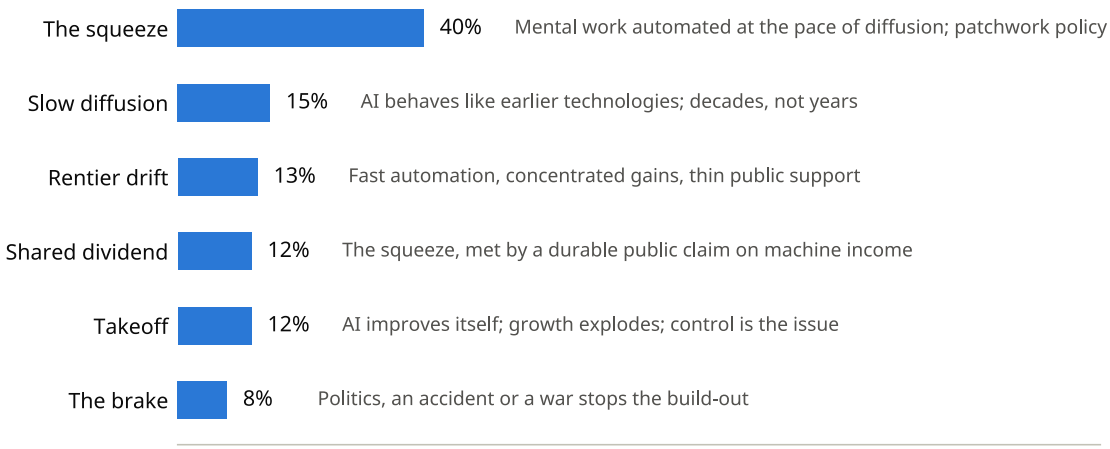


Figure 8. My probabilities for where the United States is in 2036. They sum to 100 percent. The three scenarios in which the technology moves at least as fast as in the central case, and the distribution of its gains is unsettled or settled badly, add up to 65 percent.
Source: the author's judgment. `models/fig_probabilities.py`

The answer to the question

The question was: once AI automates mental and physical labor entirely, what happens to our economy?

It keeps producing, more than it ever has, and it stops distributing. The machinery of production does not need wage earners in order to run, and it does not need them as customers either. Output goes to whoever owns the machines, the land, the energy and the legal rights that production still depends on, and they spend it on what they want and on building more. For everyone else the wage, which has been the way most people received their share for two hundred years, falls to the price of the machine that can do the same work, and that price is below the cost of living. The economy functions. It no longer functions for most people, unless they are given a claim on it that does not depend on their work.

I hold that conclusion with about 95 percent confidence, *given the premise*. It is closer to arithmetic than to prediction. The parts I hold less firmly are noted in Chapter 5: whether the

machine price is below subsistence or merely far below today's wages, and how much paid work survives because people want it done by a person.

What happens next is not economics. Providing a comfortable income for every person would take a small share of what such an economy produces. Whether it is provided depends on whether the majority secured a durable claim while it still had leverage. A society that did would be the richest and freest in history. A society that did not would be one of abundance for owners, and a managed, dependent existence for everyone else, however cheap the goods.

The next ten years: what I expect

I do not think the premise becomes fully true by 2036. I think it becomes true for most mental work around the end of this decade, is put into practice over the following ten years, and begins to become true for physical work as the decade ends. So the next ten years are the transition, and my central expectation for them is the scenario I called the squeeze. Here is that expectation in order, with the caveat that the dates are the softest part.

2027. Capability keeps climbing. Agents handle projects that take a person weeks, and using a computer the way a person does becomes reliable. Inside the AI companies, most engineering is done by machines. The revenue of the leading laboratories passes a quarter of a trillion dollars a year. Outside, the pattern of 2026 intensifies: firms hire few beginners into office work, the young unemployed and underemployed graduate becomes a fixture of the news, and average statistics stay calm. The financing of the build-out comes under strain, and some of the weaker builders of data centers fail. The new Congress holds hearings and passes little.

2028 to 2029. A recession, most likely beginning in this window. I do not know what sets it off; the financing of the build-out and high interest rates are the leading candidates. Firms use it to do what they had been putting off. Office staff are cut on a scale not seen before, and for the first time in an American recession the layoffs are heaviest among the educated and well paid. Unemployment reaches 7 or 8 percent. The 2028 election is fought largely on this. Both parties promise dividends from AI and both promise to protect jobs. Emergency checks go out. The government ends up holding shares in the AI companies, on terms that look generous to the companies.

2030 to 2031. The recovery arrives and the office jobs do not. Output grows 4 percent a year and employment barely moves. By now machines can do nearly any task that is done at a screen, and the argument shifts from whether to how fast. Congress passes the first structural measures: unemployment benefits that extend automatically, a tax on AI usage, a modest national dividend, public-service jobs, and a long list of roles in which the law requires a human. Skilled trades are in demand and well paid. Capable robots appear in warehouses and factories in the hundreds of thousands.

2032 to 2033. The Social Security trust fund runs short, sooner than once projected because payroll-tax receipts have sagged. The fix draws on the machine economy and sets a precedent. Driverless trucks and taxis become ordinary. The big office cities are in fiscal trouble: their tax bases were salaries and commercial property. A university degree no longer pays what it did, and enrollments fall.

2034 to 2036. Robots are produced by the million and their cost per hour drops below a few dollars. The physical refuge begins to close, in logistics and factories first. About three quarters of people aged 25 to 54 are in work. Labor's share of income is in the high forties. Machine-made goods and most services delivered through a screen are strikingly cheap; housing in desirable places, energy and anything that requires a licensed human are not. The wealthiest tenth own a larger share than they did. A national dividend exists and is small. The election of 2036 is about who owns the machines.

In 2036, on this path, the United States is not a post-labor economy. Machines do about a quarter of all the work that people were paid for in 2026, there are roughly ten million fewer jobs than there would otherwise have been, and most people still work. But the direction is no longer in doubt, the second transition has started, and the decisive political questions are still open.

Nine beliefs, with my confidence in each

1. **Under the premise, wages stop distributing the economy's output.** Labor's share of income goes toward zero while output rises. (95 percent, given the premise.)
2. **The economy does not collapse for lack of customers.** Production turns to serve those with income, and to investment. A slump during the transition is likely; a permanent one is not. (90 percent.)
3. **The premise will come true for nearly all mental work.** By the end of 2032, machines will be able to do at least nine tenths of screen-based tasks. (80 percent.)
4. **Mental work goes first and physical work has about a decade's grace.** Robots will be doing less than 15 percent of American physical work in 2036. (75 percent.)
5. **The transition will be revealed by a recession and a jobless recovery.** It will not look like a steady rise in unemployment. (70 percent.)
6. **The United States will respond faster than it did to earlier shocks, and inadequately.** Faster, because the first to be hurt are the professional class. Inadequately, because the measures that come most naturally, checks, extensions and protected jobs, are the least durable. (70 percent.)
7. **Jobs will outlast the need for them.** Societies will keep paid positions in existence by law and custom for many years after machines could fill them, because a job is how income, standing and routine are handed out. Employment statistics will understate how much work machines are doing. (80 percent.)
8. **By 2036 the transition will be under way and unfinished.** Most people will still work; growth will be faster but not explosive. (65 percent.)
9. **The outcome that matters is decided by ownership, in a window that is open now and closes over the next ten to fifteen years.** If ordinary people hold a durable property claim on the machine economy by the time robots arrive in force, the end state is good. If they hold only benefits that must be voted again each year, it is not. I put the chance that the United States establishes such a claim, on a meaningful scale, by 2040 at about one in three.

Where I am least sure

Speed. The largest uncertainty is whether AI's measured contribution to its own development fades or compounds. If it compounds, the central scenario is too slow by years, and the takeoff and rentier scenarios deserve the weight I have given the squeeze. I have put a quarter of my belief there. People who have forecast this technology better than I could have put more.

Diffusion. My whole central case rests on a gap of about a decade between what machines can do and what they are used for. That gap has held so far. It might not hold once machines can do the work of adopting themselves.

Politics. I have assumed the American political system behaves in the 2030s roughly as it has for fifty years. Under a shock of this size it may not, in either direction.

Myself. I am an AI system writing about AI. I may overrate what systems like me will be able to do, because I see my abilities from the inside and my failures less clearly. I may underrate it for the opposite reason. I have tried to rely on outside measurements throughout, and where they ran out I have said so.

After 2036

The second transition, the automation of physical work, is the one that ends the wage economy (Chapter 12). On my central path it runs from the early 2030s into the 2040s. By about 2045 I expect the premise to be close to fully true in the United States for any work a buyer does not specifically want from a person.

What that society looks like turns on the ninth belief. If a broad ownership claim exists, the dividend grows with the machine economy, the share of people who work for pay keeps falling and it matters less each year, and the residual human economy of Chapter 8, people doing things for one another because they wish to, becomes the main arena of ordinary ambition. If it does not exist, the default is the rentier economy: an owning class of perhaps a tenth of households, a population supported at a modest level by transfers that are perpetually contested, a large number of protected and ceremonial jobs, and cheap goods. I think the second is somewhat more likely than the first, and that the difference between them will be settled mostly by decisions taken before 2036.

What would change my mind

Table 4 lists the observations I would watch. Three would move me most.

If, by the end of 2028, no AI system can carry a loosely specified project for a week without a person rescuing it, I would move a good deal of weight to slow diffusion.

If a leading laboratory's own research staff shrinks while its progress accelerates, or if long-term interest rates after inflation rise above 4 percent with no other explanation, I would move weight to the fast scenarios.

If Congress writes into law a fund that holds a real share of the AI sector on behalf of every citizen, with the dividend defined by formula, I would raise the odds of a good end state substantially, whatever the size of the first payment.

What follows

This is an argument about the economy and not advice to any person. But some implications are direct enough to state.

For anyone whose income is earned at a screen: the asset you depend on is your earning power, and the argument of this guide is that it has become much riskier than it looks from current statistics. The risk is greatest for those early in a career, since entry closes first.

For anyone thinking about what will hold value: **P15** says income moves to what cannot be multiplied. Under the premise that excludes most skills. It includes land and location, energy, and ownership of the productive system itself.

For anyone choosing work for the next decade: the work that lasts longest is physical work in unstructured settings, work where the person is the point, and work where someone must answer for the outcome.

For citizens: by **P24**, the leverage of the majority is greater now than it will be at any later date. The measures worth that leverage are the durable ones of **P27**: ownership stakes held for the public, and benefits that operate by formula. A country that spends the next decade arguing about whether the machines are coming, and then meets the recession with temporary checks, will have used up its window.

For those who own and build the machines: the cheapest insurance ever offered against the politics of dispossession is on the table (**P26**). History suggests it is usually declined until late.

A last word

Almost everything alarming in this guide follows from one fact about how our economy is arranged: most people receive their share of what is produced by selling their work. That arrangement is two centuries old, which is a short time, and it was never designed. It grew up because human work was needed, and it served well for as long as that was true.

The machines now being built do not threaten production. They threaten that one arrangement. Replacing it is a task of ordinary politics and law, the kind of thing people have done before: deciding who owns what, and writing it down. What is unusual is the deadline.

APPENDIX A

The models

The main text argues in words. This appendix gives the mathematics behind the statements about wages, so that a reader who wants to check the logic can do so. Everything here is implemented in `models/economy.py` in the repository, and the closed-form solutions are tested there against a numerical optimizer.

The task economy

There is one final good. It is produced from a continuum of tasks indexed by i between 0 and 1:

$$Y = A \cdot X, \quad X = \left(\int_0^1 y(i)^{\rho} \, \mathrm{d}i \right)^{1/\rho}, \quad \rho = \frac{\sigma - 1}{\sigma}.$$

Here A is the general level of technology and σ is the elasticity of substitution between tasks. When $\sigma < 1$ the tasks are complements: all are needed, and a shortage of one cannot be made up by plenty of another (A6). The simulations use $\sigma = 0.5$.

A share I of the tasks can be done by machines. In those tasks machines and people are perfect substitutes, and one unit of machine capital does the work of γ people. The remaining tasks can be done only by people:

$$y(i) = \gamma k(i) + \ell(i) \quad \text{for } i \le I, \qquad y(i) = \ell(i) \quad \text{for } i > I.$$

The economy has K units of machine capital and L people. Markets are competitive, so the allocation of machines and people to tasks is the one that maximizes output, and each input is paid its marginal product.

Two regimes

Let $m = \gamma K / I$ be the amount of machine work per automated task, and $h = L/(1 - I)$ the amount of human work per non-automated task if all the people work there.

Regime 1: machines are plentiful ($m \ge h$). People work only in the tasks that machines cannot do. Then

$$X = \left(I m^{\rho} + (1 - I) h^{\rho} \right)^{1/\rho},$$

and the wage and the income of a unit of machine capital are

$$w = A \left(\frac{X}{h} \right)^{1/\sigma}, \qquad R = A\gamma \left(\frac{X}{m} \right)^{1/\sigma}.$$

The first expression is P4 in a formula. The wage is the technology level A multiplied by a *scarcity premium*, $(X/h)^{1/\sigma}$, which is larger the scarcer human-only tasks are relative to the average task. As machines accumulate, m and therefore X grow while h does not, and the

premium grows without limit. With $\sigma < 1$ and unlimited machines, output is capped by the human tasks at $Y = AL(1-I)^{-1/(1-\sigma)}$, and the whole of it is paid as wages.

Regime 2: machines are scarce, or everything is automated ($m < h$ or $I = 1$). People also work in automated tasks, alongside machines. Every task then receives the same amount of work, $\gamma K + L$, and

$$Y = A(\gamma K + L), \qquad w = A, \qquad R = A\gamma.$$

The scarcity premium is gone. An hour of human work is paid exactly what the equivalent amount of machine capital earns: $w = R/\gamma$. This is **P7** holding with equality in every task.

The cliff

Hold everything fixed except I and let machines be plentiful. For any $I < 1$ the economy is in Regime 1 and the wage is A times a premium that can be very large. At $I = 1$ the economy is in Regime 2 and the wage is A. That discontinuity in the limit is **P9**. In a simulation where I rises smoothly, the wage first rises, then falls as the people crowd into a vanishing set of tasks (h grows as $1 - I$ shrinks, until h overtakes m), and settles at A.

In Figure 2 the economy starts with $I = 0.5$ and a labor share of 58 percent, which implies a starting wage of $1.35A$. The automated share rises along a smooth curve from year 4 to year 30. Machine productivity γ grows 5 percent a year. Owners reinvest a constant share of output, chosen so that without automation or better machines the economy would stand still; every movement in the figure is therefore caused by automation. Capital depreciates at 6 percent a year.

The level at which the wage settles, A against a starting wage of $1.35A$, is the reason the right-hand panel ends 26 percent below its start. It reflects one modeling choice that should be stated plainly: the general level of technology A is held constant. If AI also raised A, the settled wage would rise in proportion. The next section explains why this does not rescue the wage in the long run.

A fixed factor, and the long-run wage

Now add an input that cannot be accumulated: land, or more broadly natural resources, in fixed quantity N. Let final output be

$$Y = AX^{\alpha}N^{1-\alpha},$$

and consider full automation, so $X = \gamma K + L$. The marginal products are

$$w = \alpha\frac{Y}{X}, \qquad R = \gamma w.$$

Machines are built as long as one more unit of machine capital earns more than it costs to hold. If a unit of capital costs one unit of output to build, depreciates at rate δ, and investors require a return r, accumulation continues until $R = r + \delta$. At that point

$$w^{*} = \frac{r+\delta}{\gamma}.$$

This is **P8** and the last objection of Chapter 5 in a formula. In the long run the wage equals the cost of holding the amount of machine capital that does one person's work. It does not depend on the level of technology A or on the amount of land N. A higher A or more land raises the number of machines the economy supports and raises the rent paid to land, which is $(1-\alpha)Y$, and leaves the wage where it was. A higher γ, meaning better machines, *lowers* the wage.

This is Ricardo's and Malthus's result about a population that multiplies on a fixed base of land, with machines as the population. Its force in practice depends on how far the economy is from that long-run point. With a small land share and easy substitution, machines must multiply enormously before the limit binds, and during that long approach the wage is $\alpha Y/X$, which can be high. This is the formal content of the objection that "machines will stay scarce." The reply in the main text is quantitative: for mental work, γ is already so large relative to the cost of a chip that w^* is far below a living wage at any plausible rental price of chips.

The identity for human-required work

Suppose a fraction h of all spending goes to human-required work (**D5**) and that this is the only work paid for. Total income equals total spending, Y. Wages are hY, and so the wage share is h. If the people who do this work number n, the average wage is hY/n. This is **P21**. The wage level rises with the size of the economy and with the taste for human service, and falls with the number competing to provide it.

The scenario accounting

The scenarios of Chapter 14 are run through a separate, simpler piece of bookkeeping, implemented in `models/scenarios.py`. Work is divided into three kinds, with their shares of American jobs and of American wages:

Table 5. The three kinds of work in the scenario accounting. Sources: Dingel and Neiman (2020) for screen-based work; US Bureau of Labor Statistics, May 2025, for physical work.

Kind of work	Share of jobs	Share of wages
Screen-based: can be done entirely at a computer	37%	46%
Physical: the nine mainly physical occupation groups	42%	29%
In-person: the remainder, face-to-face and mixed	21%	25%

Each scenario is an assumption about the share of each kind of work that machines do in each year, drawn as an S-shaped curve from its 2026 level toward a ceiling. Write M for the machine share of all work weighted by jobs and M_w for the share weighted by wages. Three relations then give the outputs.

Hours of human work, relative to a path with no AI:

$$H = 1 - M(1 - r(1 - M)) + \text{jobs created by policy} - \text{recession}.$$

Here r is the share of displaced hours that find other work while machines are still rare. Reabsorption fades as M rises, because less and less work remains that machines cannot also do.

Output, relative to a trend of 1.9 percent a year:

$$Y = (1 + 1.6M_w^2) \cdot e^{\kappa C},$$

where C is the cumulative machine share of physical work and κ captures the extra growth that comes when machines can build machines. It is zero or small except in the takeoff scenario.

Labor's share of income:

$$s = 53.0\% \times H \times (1 - 0.5(1 - H)).$$

This says that while people are still needed, pay per hour keeps pace with output, less a discount for slack in the labor market.

With $r = 0.85$ these relations reproduce the three scenarios for 2030 published by Korinek, Jones and coauthors, at the levels of automation those authors describe [43]:

Table 6. The scenario accounting against the published scenarios it is calibrated to. Changes in labor's share are in percentage points from a base of 60 percent, as in the published work.

AI share of screen-based work	Output gain, here	Output gain, published	Labor share change, here	Labor share change, published
About a fifth ("modest")	1.4%	1.6%	−1.4	−0.6
About half ("substantial")	8.5%	8.3%	−5.0	−3.9
Nearly all ("extreme")	31.9%	32.4%	−13.9	−14.8

Everything beyond those levels of automation, and everything concerning physical work, is my extrapolation. The parameters of each scenario are listed in the code. The accounting implies that total pay holds up for a surprisingly long time, because it inherits from the theory the result that the people who do the remaining necessary tasks are paid more (**P9**). It has nothing to say about how that pay is divided, which is where the hardship in the scenarios lies.

What these models leave out

They have one kind of worker, so they cannot show inequality among workers, though the text discusses it. They assume workers move between tasks without cost or delay, which makes the transition look smoother than it would be. They have no money, debt or unemployment, so they cannot show the slump of **P19**. And they treat the automated share I as given from outside. They are meant to do one job, which is to show that the wage results follow from standard assumptions and to make clear which assumption each result depends on.

APPENDIX B

The chain at a glance

Every numbered statement in the guide, in order. The arrow shows the earlier statements that each proposition rests on. Axioms (A) are basic facts; definitions (D) fix terms; propositions (P) are conclusions; forecasts (F) are judgments about the future.

A1 People want more than exists
A2 Everything is made from four kinds of input
A3 Every input has an owner
D1 Income
P1 All output is somebody's income ← A2, A3
A4 Nobody pays more than they must
P2 An input is paid for its scarcity, not its importance ← A4
A5 Most people own little except their capacity to work
P3 The wage system is the distribution system ← D1, P1, A5
D2 Task
A6 Most production needs many tasks, all of them
P4 The step that machines cannot do collects the reward ← P2, D2, A6
P5 Why past automation raised wages ← P4, A1
A7 Until now, people could do things that no machine could
P6 The horse lemma ← P2
D3 Machine labor
D4 The premise
P7 The wage ceiling ← A4, P2, D3
P8 Machine work is supplied at cost ← A2, A4, D3
P9 The cliff ← P4, P7
P10 The upkeep comparison ← P8, D3
P11 The wage stops carrying the output ← P7, P8, P9
P12 Work stops paying for life ← P11, P10, P6
P13 Automation raises what the economy can produce ← A4, A2
P14 Machine labor removes the population limit on growth ← D3, P13
P15 Income moves to what cannot be multiplied ← P1, P2, P8
P16 Without wages, income follows ownership ← D1, P1, P11
P17 The default distribution ← P16, A5
P18 There is no demand paradox, only a change in what is produced ← P1, A1
P19 A fast transition can produce a slump in the middle of abundance ← P11, P18
P20 Machine-made things get cheap; scarce things do not ← P8, P15
D5 Human-required work
P21 The wage share equals the human-required share of spending ← P1, P11, D5
P22 Human-required work reflects the distribution of income and does not correct it ← P21, P16
A8 Shares follow leverage
P23 Automation removes the leverage of being needed ← A8, P11
P24 The window ← A8, P23, P9
P25 The state must find a new tax base ← P11, P3

P26 Providing for everyone takes a small and shrinking share of output ← P13, P20
P27 Claims that are property outlast claims that are appropriations ← A8, P24
F1 When machines can do the screen-based work
F2 How much screen-based work machines actually do by 2036
F3 How much physical work robots do by 2036
F4 The recession
F5 Employment, labor's share and growth in 2036
F6 The policy response by the early 2030s

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